Neural Networks 206 (2027) 109868

Contents lists available at ScienceDirect

Neural Networks

journal homepage: www.elsevier.com/locate/neunet

Full Length Article

Dual-space learning with low-rank and sparse constraints for multi-view subspace clustering ⋆

Lihao Yang [a], Yong Wang [a,*], Yourui Huang [b], Guifu Lu [a], Geng Huang [c]

[a] School of Computer Science and Information, Anhui Polytechnic University, Wuhu, Anhui, 241000, China
[b] School of Electrical Engineering, Anhui Polytechnic University, Wuhu, Anhui, 241000, China
[c] School of Computer Science and Technology, University of Science and Technology of China, Hefei, Anhui, 230000, China

ARTICLE INFO

Keywords:
Multi-view clustering low-rank representation
Sparse subspace learning
Latent representation

ABSTRACT

Multi-view subspace clustering (MSC) has attracted considerable attention because it integrates consistent and complementary information from multiple views. However, many existing methods learn different types of structural information within a single representation space, which may not clearly distinguish their respective roles. First, representing heterogeneous information from different views within a single space may make it difficult to preserve both global consistency across views and local structures within individual views. Second, although low-rank and sparse regularizations have been widely used in subspace clustering, their roles in different latent representation spaces have received less attention. To address these issues, we propose Dual-Space Learning with Low-Rank and Sparse Constraints for Multi-View Subspace Clustering (DSLSMSC). Specifically, the proposed method simultaneously learns latent representations from the joint-view space and the original-view spaces. Low-rank regularization is imposed on the joint-view representation to capture the shared structure across views, while sparse regularization is imposed on the original-view representations to preserve view-dependent local relationships. The learned representations are further integrated via tensor nuclear norm (TNN) regularization to exploit high-order correlations across spaces. Experimental results on multiple benchmark datasets demonstrate the effectiveness of DSLSMSC.

1. Introduction

Modern data are often collected from multiple sources, and each view provides only a partial description of the samples. Consequently, a single view may not capture all relevant structural information. Multi-view data have therefore become an important topic in machine learning. For example, web and multimedia data may describe the same sample through visual features, textual descriptions, and hyperlinks. Nevertheless, the effective clustering of such heterogeneous data remains a significant challenge. Multi-view clustering (MVC) (Chao et al., 2021) integrates consistent and complementary information across views and has been applied in areas such as medical diagnosis and social network analysis.

Compared with traditional clustering algorithms such as k-means (MacQueen, 1967), MVC aims to learn an affinity matrix that better reflects the intrinsic relationships among samples by integrating complementary information from multiple views. Existing MVC methods can be broadly grouped into spectral clustering (Cai et al., 2024; Xie et al., 2024), non-negative matrix factorization (NMF) (Li et al., 2023, 2025), subspace learning (He et al., 2025; Xie et al., 2023), and deep learning-based methods (Cui et al., 2024; Gao et al., 2023). Spectral clustering based methods mainly rely on graph construction to describe sample affinities across views, whereas NMF-based methods focus on learning compact low-dimensional representations with part-based interpretability. Deep learning methods use nonlinear transformations and hierarchical feature extraction to obtain informative representations. In contrast, subspace methods describe relationships among samples through self representation and are well suited to identifying latent subspace structures in high dimensional multi-view data. For this reason, subspace-based approaches have attracted considerable attention in recent years.

MSC assumes that each sample can be expressed as a linear combination of other samples from the same subspace. Huang et al. (2022) proposed MSC by joint measuring of consistency and diversity, which jointly exploits consistency and diversity information across multiple views to improve clustering performance. Yu et al. (2025) presented robust multi-view clustering with hyper-Laplacian regularization, which

⋆ This work was supported by the Wuhu "ChiZhu Light" Major Science and Technology Project (2023zc10).
* Corresponding author.
E-mail addresses: 2240932113@stu.ahpu.edu.cn (L. Yang), yongwang@ahpu.edu.cn (Y. Wang), huangyourui@ahpu.edu.cn (Y. Huang), lu-guifu@ahpu.edu.cn (G. Lu), hgeng@mail.ustc.edu.cn (G. Huang).

https://doi.org/10.1016/j.neunet.2026.109868
Received 19 May 2025; Received in revised form 13 July 2026; Accepted 2 November 2026
Available online 5 November 2026
0893-6080/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

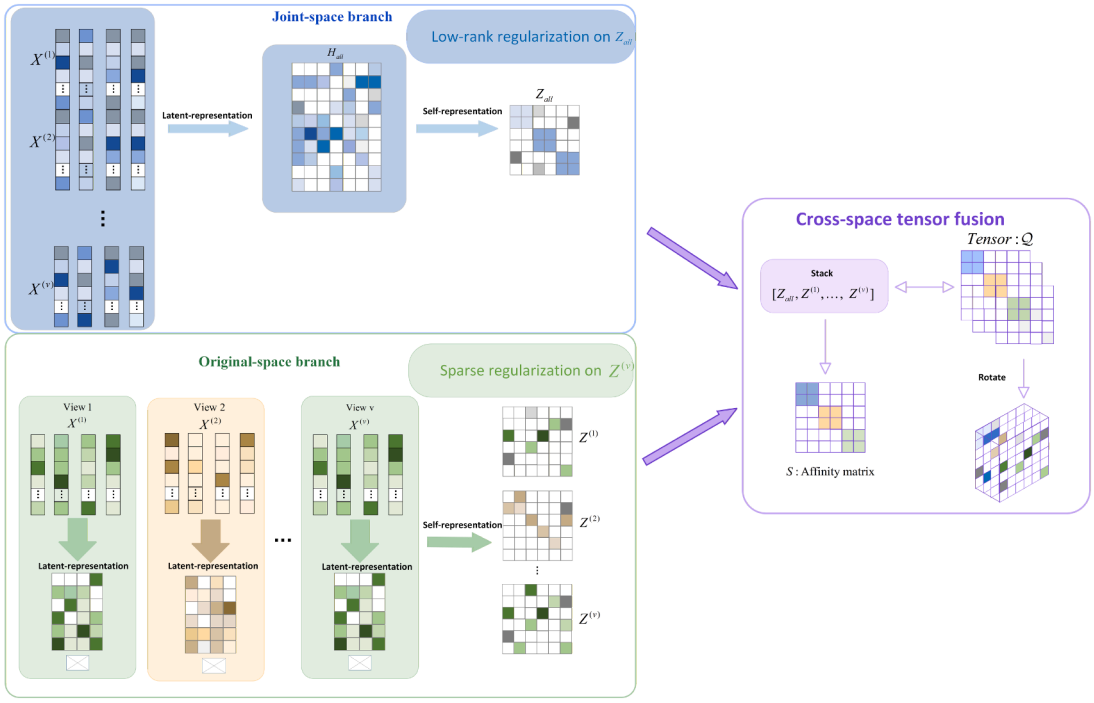


Fig. 1. The flowchart of DSLSMSC.

enhances clustering robustness by preserving the intrinsic geometric structure of multi-view data through hyper-Laplacian regularization. Luo et al. (2025) proposed Multi-view subspace clustering via slack consistency and double-side orthogonal diversity, which introduces slack consistency matrices and double-side orthogonal diversity to better balance consistency and complementarity across multiple views.

However, many existing MSC methods still rely on representation learning in the original feature space, where redundant features may degrade the quality of the learned subspace representation. In contrast, latent-space learning can often provide more compact and discriminative representations for clustering. Zhang et al. (2017) proposed Latent Multi-view Subspace Clustering (LMSC), which projects multi-view data into a common latent space to integrate complementary information. MCLES (Chen et al., 2020) learns a latent embedding from multi-view features and constructs similarity matrices from the resulting embeddings, reducing the influence of corrupted observations. However, two issues remain. First, although latent representations can improve compactness, placing heterogeneous information in a single space may obscure the differences between global and view dependent structures. Second, although low-rank and sparse regularizations have both been widely used in multi-view subspace clustering, explicitly assigning them to different latent representation spaces according to their structural roles remains less explored.

To address the above issues, we propose a novel Dual-Space Learning with Low-Rank and Sparse Constraints for Multi-View Subspace Clustering (DSLSMSC) method. As illustrated in Fig. 1, DSLSMSC consists of three key steps. First, we construct a joint-view feature representation via feature concatenation to expose complementary cross-view information, and then simultaneously learn latent self-representation matrices in both the joint-view space and the original-view spaces. The joint view and original view branches are assigned different regularizations according to the structural information retained in each space. Specifically, the joint-view branch and the original-view branches are assigned different structural priors according to their modeling roles. Second, low-rank regularization is imposed on the joint-view self-representation to capture global shared structure, while sparse regularization is imposed on the original-view self-representations to preserve local view-specific relationships. Third, a TNN constraint is introduced to couple the learned representations and capture their high-order cross-space correlations. Finally, an efficient optimization algorithm based on the augmented Lagrangian and alternating direction minimization framework is developed to solve the resulting model. Extensive experiments show that DSLSMSC achieves favorable clustering results across multiple benchmark datasets.

The contributions of this work can be summarized as follows:

- We propose a dual space latent self representation framework that learns representations from a joint view space and multiple original view spaces, allowing shared information across views and local relationships within individual views to be considered simultaneously.
- We design a dual space regularization strategy, in which low rank regularization is imposed on the joint view self representation to capture shared global structure, sparse regularization is imposed on the original view self representations to preserve local discriminability, and TNN regularization is used to integrate the learned representations and exploit their high order correlations.
- We develop an iterative optimization algorithm based on the ADMM framework to efficiently solve the DSLSMSC optimization problem, with comprehensive experimental validation across multiple benchmark datasets demonstrating its effectiveness and competitive performance.

2. Related work

Current research on constraint design for representation matrices can be roughly categorized into single-prior and hybrid-prior approaches. MLRR (Chen et al., 2021) extends LRR by imposing symmetry constraints on the affinity matrix, aligning row-wise and column-wise information to reinforce its block-diagonal structure. To address the limitation of treating all features equally, Fu et al. (2022) introduces an auto-weighted penalty mechanism that emphasizes informative features while suppressing noisy ones. Zhang et al. (2023) decomposes view-specific representation matrices into two smaller factor matrices,

thereby reducing the computational cost of SVD while enforcing low-rank constraints on feature matrices. These methods demonstrate the effectiveness of low-rank or sparse priors for representation learning. In addition, hybrid sparse low-rank constraints have been widely investigated in subspace clustering and multi-view clustering, where low-rank and sparse priors are jointly used to provide complementary structural information for affinity matrix construction. For instance, Brbić and Kopriva (2018) simultaneously imposes both low-rank and sparsity constraints on feature matrices to construct more expressive affinity matrices. A joint NMF framework (Liu et al., 2013) combines low-rank and sparse constraints to address consensus learning, noise robustness, and structure preservation in multi-view clustering. Furthermore, Nie et al. (2018) integrates Procrustes-based alignment with low-rank and sparse constraints to ensure both geometric consistency and discriminative feature learning. These studies indicate that joint sparse low-rank modeling is a valid and effective strategy. Current research on constraint design for representation matrices can be categorized into single-constraint and hybrid-constraint approaches. MLRR (Chen et al., 2021) extends LRR by imposing symmetry constraints on the affinity matrix, aligning row-wise and column-wise information to reinforce its block-diagonal structure. To address LRR's limitation of treating all features equally. Fu et al. (2022) introduces an auto-weighted penalty mechanism that emphasizes informative features while suppressing noisy ones. Zhang et al. (2023) decomposes view-specific representation matrices into two smaller factor matrices, simultaneously reducing SVD computational overhead and enforcing low-rank constraints on feature matrices. However, these approaches typically employ either low-rank or sparse constraints alone, resulting in incomplete feature representation. Recent studies have shifted toward joint constraint optimization. For instance, Brbić and Kopriva (2018) simultaneously imposes both low-rank and sparsity constraints on feature matrices to construct more expressive affinity matrices. A joint NMF framework (Liu et al., 2013) that combines low-rank and sparse constraints has been developed to address three critical aspects in multi-view clustering: consensus learning, noise robustness, and structure preservation. Furthermore, Nie et al. (2018) integrates Procrustes-based alignment with low-rank and sparse constraints to ensure both geometric consistency and discriminative feature learning.

Compared to self-representation matrix-based subspace clustering, latent representation-based approaches can achieve more powerful feature representations for enhanced clustering performance. Zhao and Lu (2022) innovatively applied Robust Principal Component Analysis (RPCA) to multi-view learning by constructing a shared latent subspace and jointly optimizing low-rank constraints with sparse error terms, thereby obtaining cleaner and more robust affinity matrices. Zheng et al. (2023) made groundbreaking progress through dual-path learning of latent feature representations and subspace representations, simultaneously exploiting available information in both representation spaces to fully explore the consistency and complementarity of multi-view data during learning. Lan et al. (2024) developed a hierarchical strategy that processes view-specific correlations at lower levels while exploring view-consistent correlations at higher levels, effectively mining feature-space diversity in early stages and consistency in advanced stages.

In recent years, tensor-based multi-view clustering methods have emerged as a research hotspot due to their unique capability for modeling high-order relationships. Wang et al. (2021) integrated probability transition matrices from all views into a third-order tensor with norm constraints on the error matrix, fully exploiting the latent value of multi-view data to significantly enhance algorithmic robustness. Xia et al. (2021) introduced a weighted tensor Schatten p-norm based on t-SVD to leverage both embedded clustering structures and high-order information across views. Wu and Lu (2024) stacked two self-expression matrices into a tensor and introduced TNN constraints, thereby enabling mutual reinforcement during learning to obtain cleaner block-diagonal representations. Sun et al. (2025) proposed JETLG, which enhances affinity graph learning by jointly modeling intra-view and inter-view structural information through tensor low-rank regularization. Tao et al. (2024) developed a tensor-based multi-view spectral clustering method via shared latent space, which effectively combines shared latent representation learning with tensor modeling to better exploit high-order correlations across view.

However, existing multi-view subspace clustering methods still face an important limitation. Although many methods employ latent representation learning to improve robustness and compactness, they often learn all structural information within a single representation space, which makes it difficult to simultaneously characterize the shared global structure across views and the view-specific local structure of each view. In addition, tensor-based methods are effective in modeling high-order correlations, but they are usually built upon representations learned in a unified manner and therefore may not fully exploit the different structural roles of different spaces. As a result, shared global structure, local view-specific structure, and high-order cross-view correlations are often not modeled in a sufficiently coordinated way within a unified framework. Motivated by this observation, we develop a dual-space framework that simultaneously learns latent representations from the joint-view space and the original-view spaces, applies low-rank and sparse regularizations to characterize their different structural properties, and further integrates them through tensor-based fusion. In this way, the proposed method provides a more comprehensive characterization of multi-view data for clustering.

3. Preliminaries

This section elaborates on the theoretical foundation of our model—the Tensor Singular Value Decomposition (t-SVD). For clarity, Table 1 summarizes the key mathematical notations and formalisms used throughout this paper. The t-SVD-based TNN, serves as the core mathematical tool in our work. To facilitate a deeper understanding of its operational mechanism, we present the following block multipliers (Kilmer & Martin, 2011):

For a tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, its block circulant matrix $\mathrm{bcirc}(\mathcal{X})$ is constructed as:

$$\mathrm{bcirc}(\mathcal{X}) = \begin{bmatrix} X^{(1)} & X^{(n3)} & \cdots & X^{(2)} \\ X^{(2)} & X^{(1)} & \cdots & X^{(3)} \\ \vdots & \vdots & \ddots & \vdots \\ X^{(n_3)} & X^{(n_3-1)} & \cdots & X^{(1)} \end{bmatrix} \tag{1}$$

The vectorization of tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ into a block vector and its inverse operation are defined as follows:

$$\mathrm{bvec}(\mathcal{X}) = \begin{bmatrix} X^{(1)} \\ X^{(2)} \\ \vdots \\ X^{(n_3)} \end{bmatrix} \tag{2}$$

$$\mathrm{bvfold}(\mathrm{bvec}(\mathcal{X})) = \mathcal{X} \tag{3}$$

Table 1
Description for main notations.

Notations	Descriptions
d	The number of features in the original view
m	The number of features in the joint view
n	The number of samples
k	The dimension of subspace
I	The identity matrix with compatible sizes
$\|X\|_*$	The trace norm of matrix X
$\|X\|_1$	The l_1-norm of matrix X
$\|X\|_{2,1}$	The $l_{2,1}$-norm of matrix X
$\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$	The third-order tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$
$\bar{\mathcal{X}} = \mathrm{fft}(\mathcal{X}, [], 3)$, $\mathcal{X} = \mathrm{ifft}(\bar{\mathcal{X}}, [], 3)$	Fourier Transform and Inverse Fourier Transform

The block diagonal matrix of tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ and its inverse operation are defined as follows:

$$\text{bdiag}(\mathcal{X}) = \begin{bmatrix} X^{(1)} & & & \\ & \ddots & & \\ & & \ddots & \\ & & & X^{(n^3)} \end{bmatrix} \tag{4}$$

$$\text{bdfold}(\text{bdiag}(\mathcal{X})) = \mathcal{X} \tag{5}$$

Additionally, several key tensor definitions are enumerated below:

Definition 1 (t-Product). Let the tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}, \mathcal{Y} \in \mathbb{R}^{n_2 \times n_4 \times n_3}$. Then the t-product of $\mathcal{X}$ and $\mathcal{Y}$ is defined as:

$$\mathcal{X} * \mathcal{Y} := \text{bvfold}\{\text{bcirc}(\mathcal{X})\text{bvec}(\mathcal{Y})\} \in \mathbb{R}^{n_1 \times n_4 \times n_3} \tag{6}$$

Definition 2 (Identity Tensor). Consider a tensor $\mathcal{I} \in \mathbb{R}^{n_1 \times n_1 \times n_3}$. It is deemed an identity tensor when its first frontal slice is an $n_1 \times n_1$ identity matrix, and the remaining frontal slices are zero matrices:

Definition 3 (Tensor Transpose). Let $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ be a third-order tensor, its transpose tensor $\mathcal{X}^{\mathrm{T}} \in \mathbb{R}^{n_2 \times n_1 \times n_3}$ is obtained by applying matrix transposition to each frontal slice from the second to the last.

Definition 4 (Orthogonal Tensor). A tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is called an orthogonal tensor if it satisfies

$$\mathcal{X}^{\mathrm{T}} * \mathcal{X} = \mathcal{X} * \mathcal{X}^{\mathrm{T}} = \mathcal{I} \tag{7}$$

where $*$ denotes the t-product operation defined in Definition 1. $\mathcal{X}^{\mathrm{T}}$ and $\mathcal{I}$ is the transpose tensor and the identity tensor of $\mathcal{X}$. For detailed explanations, refer to Definitions 3 and 2.

Definition 5 (f-Diagonal Tensor). A tensor is called an f-diagonal tensor if every frontal slice is a diagonal matrix.

Definition 6 (TNN Xie et al., 2018). Give a tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, its Tensor Nuclear Norm (TNN) is defined as:

$$\|\mathcal{X}\|_{\text{TNN}} = \sum_{i=1}^{\min(n_1, n_2)} \sum_{j=1}^{n_3} |S(i, i, j)| \tag{8}$$

Here, S is the f - diagonal tensor obtained by performing the tensor-Singular Value Decomposition(t-SVD) on the tensor $\mathcal{X}$.

4. The proposed approach and optimization

In this section, we present the DSLSMSC algorithm and first clarify the motivation of the proposed dual-space constraint allocation. DSLSMSC aims to assign different structural priors to different latent representation spaces according to their modeling roles in multi-view subspace clustering. Specifically, the joint-view space, constructed by concatenating multi-view features, aggregates complementary information from different views and is therefore more suitable for capturing global consensus structures through low-rank regularization. In contrast, the original-view spaces preserve view-dependent local relationships, where sparse regularization encourages each sample to select a small number of informative neighbors and helps maintain local discriminability. The learned joint-view and original-view representations are then coupled by tensor nuclear norm regularization to exploit high-order cross-space correlations.

4.1. Joint-space learning mechanism

Inspired by LMSC (Zheng et al., 2020a), our method constructs a joint-view representation $X_{all} \in \mathbb{R}^{m \times n}$ by concatenating the original multi-view data, then derives a more comprehensive latent embedding representation H_{all} through latent projection. To account for feature representation errors, we formulate the following:

$$X_{all} = P_{all} H_{all} + E_0 \tag{9}$$

where P_{all} denotes the projection matrix, we add a constraint as $P_{all} {P_{all}}^T = I$ to it. H_{all} represents the latent representation of X_{all} and E_0 correspond to reconstruction errors.

Then, based on the joint-view latent representation H_{all}, the objective function of subspace clustering based on sub-representations is re-specified as:

$$\begin{aligned} &\min_{Z_{all}, E_0} \|E_{all}\|_{2,1} + \lambda_1 \|Z_{all}\|_* \\ &\text{s.t. } X_{all} = P_{all} H_{all} + E_0, \quad H_{all} = H_{all} Z_{all} + E_2, \\ &E_{all} = [E_0; E_2] \text{ and } P_{all} P_{all}^T = I \end{aligned} \tag{10}$$

Here, H_{all} represents the latent representation of X_{all}, and E_{all} is the error matrix obtained by vertically concatenating the errors between the latent representation and the subspace representation. Additionally, λ_1 is a trade-off parameter that balances the reconstruction error term and the regularization term for the subspace representation. Moreover, we impose a low-rank constraint on the latent self-representation matrix Z_{all} to learn the global structural information.

4.2. Original-space learning mechanism

For the original multi-view data, let $\{X^v\} \in \mathbb{R}^{d^v \times n}$ denote the set of input matrices, where X^v represents the vth view characterized by a feature dimension d^v and n samples, providing a unique perspective of the data. By employing projection matrix P^v we obtain the latent representation H^v, which is then subjected to self-representation learning with sparse constraints to capture view-specific local structural information. E_{13} is the error matrix obtained by vertically concatenating the reconstruction errors of the latent representations and the subspace representations.

$$\begin{aligned} &\min_{Z^v, P^v, H^v, E_{13}} \|E_{13}\|_{2,1} + \lambda_2 \|Z^v\|_1 \\ &\text{s.t. } X^v = P^v H^v + E_1^v, H^v = H^v Z^v + E_3^v, \\ &\quad E_{13} = [E_1^v; E_3^v], \text{Diag}(Z^v) = 0 \text{ and } P^v {P^v}^T = I \end{aligned} \tag{11}$$

4.3. Objective function

To jointly exploit the global structure from the joint-view space and the local structure from the original-view spaces, we integrate the objective function (10) for joint-space learning with the objective function (11) for original-space learning through a tensorized formulation, thereby facilitating cross-space learning.

$$\begin{aligned} &\min_{Z_{all}, Z^v, P_{all}, P^v, H_{all}, H^v, E, \mathcal{H}} \|E\|_{2,1} + \lambda_1 \|Z_{all}\|_* + \lambda_2 \|Z^v\|_1 + \lambda_3 \|\mathcal{H}\|_{TNN} \\ &\text{s.t. } X_{all} = P_{all} H_{all} + E_0, X^v = P^v H^v + E_1^v, \\ &H_{all} = H_{all} Z_{all} + E_2, H^v = H^v Z^v + E_3^v, \\ &\mathcal{H} = \Phi(Z_{all}, Z^v), \\ &E = [E_0; E_1^v; E_2; E_3^v], \\ &P_{all} {P_{all}}^T = I, P^v {P^v}^T = I \text{ and } \text{Diag}(Z^v) = 0 \end{aligned} \tag{12}$$

where λ_1, λ_2, and λ_3 are balancing parameters, and $\Phi(\hat{A}\cdot)$, denotes the operation of stacking two matrices along the third dimension to form a tensor $\mathcal{Q} \in \mathbb{R}^{n \times n \times (v+1)}$. Building upon this, we impose a rotated tensor-based TNN constraint on the tensor $\mathcal{H}$. This approach enables cross-space interaction between the joint-view low-rank representation and the original-view sparse representations, thereby enhancing the structural characterization of multi-view data.

4.4. Optimization

To facilitate the solution, we introduce auxiliary variables $\mathcal{Q} = \mathcal{H}, J_{all} = Z_{all}$ and $Z^v = J^v$ to Eq. (12), and construct the augmented

Lagrangian function $\mathcal{L}$ as follows:

$$\begin{aligned}\mathcal{L}(Z_{all}, Z^v, P_{all}, P^v, H_{all}, H^v, E, \mathcal{Q}, J_{all}, J^v) = &\|E\|_{2,1} + \lambda_1\|J_{all}\|_* \\ &+ \lambda_2\|J^v\|_1 + \lambda_3\|\mathcal{Q}\|_{\mathrm{TNN}} \\ &+ \langle Y_1, X_{all} - P_{all}H_{all} - E_0\rangle \\ &+ \frac{\mu}{2}\|X_{all} - P_{all}H_{all} - E_0\|_F^2 \\ &+ \langle Y_2, H_{all} - H_{all}Z_{all} - E_2\rangle \\ &+ \frac{\mu}{2}\|H_{all} - H_{all}Z_{all} - E_2\|_F^2 \\ &+ \left\langle Y_3^v, X^v - P^vH^v - E_1^v\right\rangle \\ &+ \frac{\mu}{2}\|X^v - P^vH^v - E_1^v\|_F^2 \\ &+ \left\langle Y_4^v, H^v - H^vZ^v - E_3^v\right\rangle \\ &+ \frac{\mu}{2}\|H^v - H^vZ^v - E_3^v\|_F^2 \\ &+ \langle Y_5, Z_{all} - J_{all}\rangle \\ &+ \frac{\mu}{2}\|Z_{all} - J_{all}\|_F^2 \\ &+ \left\langle Y_6^v, Z^v - J^v + \mathrm{Diag}(J^v)\right\rangle \\ &+ \frac{\mu}{2}\|Z^v - J^v + \mathrm{Diag}(J^v)\|_F^2 \\ &+ \langle \mathcal{W}, \mathcal{H} - \mathcal{Q}\rangle \\ &+ \frac{\mu}{2}\|\mathcal{H} - \mathcal{Q}\|_F^2 \end{aligned} \tag{13}$$

where μ are penalty parameter, Y_1, Y_2, Y_3^v, Y_4^v, Y_5, Y_6^v and tensor $\mathcal{W}$ are the Lagrange multipliers. The optimization of DSLSMSC can be achieved using the ADMM (Lin et al., 2011), which decomposes the problem into ten subproblems. These subproblems can be optimized by fixing the other variables.

(1) update Z_{all}: By fixing all other variables except Z_{all}, Eq. (13) can be expressed as:

$$\begin{aligned}\min_{Z_{all}} \quad &\langle Y_2,\ H_{all} - H_{all}Z_{all} - E_2\rangle + \frac{\mu}{2}\left\|H_{all} - H_{all}Z_{all} - E_2\right\|_F^2 \\ &+ \langle Y_5,\ Z_{all} - J_{all}\rangle + \frac{\mu}{2}\left\|Z_{all} - J_{all}\right\|_F^2 \\ &+ \langle \mathcal{W},\ \mathcal{H} - \mathcal{Q}\rangle + \frac{\mu}{2}\|\mathcal{H} - \mathcal{Q}\|_F^2 \end{aligned} \tag{14}$$

Taking the derivative of Z_{all} directly and making it zero gives a closed solution:

$$\begin{aligned}Z_{\mathrm{all}}^* = &(\mu H_{\mathrm{all}}^T H_{\mathrm{all}} + 2\mu I)^{-1}\left(H_{\mathrm{all}}^T Y_2 + \mu H_{\mathrm{all}}^T H_{\mathrm{all}} - \mu H_{\mathrm{all}}^T E_2 - W^1\right. \\ &\left. + \mu Q^1 - Y_5 + \mu J_{\mathrm{all}}\right). \end{aligned} \tag{15}$$

(2) update Z^v: By fixing all other variables except Z^v, Eq. (13) can be expressed as:

$$\begin{aligned}\min_{Z^v} \quad &\left\langle Y_4^v, H^v - H^vZ^v - E_3^v\right\rangle + \frac{\mu}{2}\left\|H^v - H^vZ^v - E_3^v\right\|_F^2 \\ &+ \left\langle Y_6^v, Z^v - J^v + \mathrm{Diag}(J^v)\right\rangle + \frac{\mu}{2}\|Z^v - J^v + \mathrm{Diag}(J^v)\|_F^2 \\ &+ \langle \mathcal{W}, \mathcal{H} - \mathcal{Q}\rangle + \frac{\mu}{2}\|\mathcal{H} - \mathcal{Q}\|_F^2 \end{aligned} \tag{16}$$

Taking the derivative of Z^v directly and making it zero gives a closed solution:

$$\begin{aligned}Z^{v*} = &(\mu(H^v)^T H^v + 2\mu I)^{-1} \\ &\left((H^v)^T Y_4^v + \mu(H^v)^T H^v - \mu(H^v)^T E_3^v\right. \\ &\left. - W^{v+1} + \mu Q^{v+1} - Y_6^v + \mu J^v\right). \end{aligned} \tag{17}$$

(3) update P_{all}: By fixing all other variables except P_{all}, Eq. (13) can be expressed as:

$$\min_{P_{all}} \frac{\mu}{2}\left\|(X_{all} + \frac{Y_1}{\mu} - E_0) - P_{all}H_{all}\right\|_F^2 \tag{18}$$

This is an orthogonal Procrustes problem, and it can be efficiently solved by utilizing Theorem 1 from Zhang et al. (2017).

(4) update P^v: By fixing all other variables except P^v, Eq. (13) can be expressed as:

$$\min_{P^v} \frac{\mu}{2}\left\|(X^v + \frac{Y_3{}^v}{\mu} - E_1{}^v) - P^vH^v\right\|_F^2 \tag{19}$$

This subproblem can be solved in a manner similar to the solution approach for the subproblem involving P_{all}, using identical methodology.

(5) update H_{all}: By fixing all other variables except H_{all}, Eq. (13) can be expressed as:

$$\begin{aligned}\min_{H_{all}} \quad &\langle Y_1, X_{all} - P_{all}H_{all} - E_0\rangle + \frac{\mu}{2}\left\|X_{all} - P_{all}H_{all} - E_0\right\|_F^2 \\ &+ \langle Y_2, H_{all} - H_{all}Z_{all} - E_2\rangle + \frac{\mu}{2}\left\|H_{all} - H_{all}Z_{all} - E_2\right\|_F^2 \end{aligned} \tag{20}$$

Taking the derivative of H_{all} directly and making it zero gives a closed solution:

$$\begin{aligned}&(\mu P_{all}{}^T P_{all} + \mu)H_{all} + \mu H_{all}(Z_{all}Z_{all}{}^T - Z_{all}{}^T - Z_{all}) \\ &= P_{all}{}^T Y_1 + \mu P_{all}{}^T X_{all} - \mu P_{all}{}^T E_0 - Y_2 + Y_2 Z_{all}{}^T + \mu E_2 - \mu E_2 Z_{all}{}^T \end{aligned} \tag{21}$$

Eq. (21) is efficiently solved by leveraging the Sylvester function available in MATLAB.

(6) update H^v: By fixing all other variables except H^v, Eq. (13) can be expressed as:

$$\begin{aligned}&(\mu P^{vT} P^v + \mu I)H^v + \mu H^v(Z^v Z^{vT} - Z^{vT} - Z^v) \\ &= P^{vT} Y_3{}^v + \mu P^{vT} X^v - \mu P^{vT} E_1{}^v - Y_4{}^v + Y_4{}^v Z^{vT} + \mu E_3{}^v - \mu E_3{}^v Z^{vT} \end{aligned} \tag{22}$$

This subproblem can be solved in a manner similar to the solution approach for the subproblem involving H^v, using identical methodology.

(7) update E: By fixing all other variables except E, Eq. (13) can be expressed as:

$$\min_{\mathrm{E}} \|\mathrm{E}\|_{2,1} + \frac{\mu}{2}\left\|E - cat(A_0, \{A_1{}^v\}_{i=1}^v, A_2, \{A_3{}^v\}_{i=1}^v)\right\|_{\mathrm{F}}^2 \tag{23}$$

where A_0, $A_1{}^v$, A_2 and $A_3{}^v$ are defined as follows:

$$\begin{aligned}A_0 &= X_{all} - P_{all}H_{all} + \frac{Y_1}{\mu}, \\ A_1{}^v &= X^v - P^vH^v + \frac{Y_3{}^v}{\mu}, \\ A_2 &= H_{all} - H_{all}Z_{all} + \frac{Y_2}{\mu}, \\ A_3{}^v &= H^v - H^vZ^v + \frac{Y_4{}^v}{\mu} \end{aligned} \tag{24}$$

and it can be effectively solved with the closed-form solution (Liu et al., 2012). Once E is optimized, E_0, $E_1{}^v$, E_2 and $E_3{}^v$ can be calculated automatically.

(8) update $\mathcal{Q}$: By fixing all other variables except $\mathcal{Q}$, Eq. (13) can be expressed as:

$$\min_{\mathrm{Q}} \frac{\lambda_3}{\mu}\|\mathcal{Q}\|_{\mathrm{TNN}} + \frac{1}{2}\left\|\mathcal{Q} - (\mathcal{H} + \frac{\mathcal{W}}{\mu})\right\|_F^2 \tag{25}$$

This subproblem related to the t-SVD-based TNN can be solved using Theorem 2 in Xie et al. (2018).

(9) update J_{all}: By fixing all other variables except J_{all}, Eq. (13) can be expressed as:

$$\min_{J_{\mathrm{all}}} \frac{\lambda_1}{\mu}\left\|J_{all}\right\|_* + \frac{1}{2}\left\|J_{all} - (Z_{all} + \frac{Y_5}{\mu})\right\|_{\mathrm{F}}^2 \tag{26}$$

By solving Eq. (26), the optimization can be performed using the singular value thresholding (SVT) method (Lin et al., 2011)

(10) update J^v: By fixing all other variables except J^v, Eq. (13) can be expressed as:

$$\min_{J^v} \frac{\lambda_2}{\mu}\|J^v\|_1 + \frac{1}{2}\left\|J^v - (Z^v + \frac{Y_6{}^v}{\mu})\right\|_{\mathrm{F}}^2 \tag{27}$$

Then, this subproblem has a closed-form solution and can be given by

$$J^{v*} = J^v - \mathrm{Diag}(J^v) \tag{28}$$

(11) update multipliers and penalty parameter: The multipliers Y_1,Y_2,Y_3^v, $\{Y_4^v\}$, Y_5, Y_6^v, $\mathcal{W}$ and the penalty parameter μ should be updated in each iteration:

$$\begin{cases} Y_1^* = Y_1 + \mu(X_{\text{all}} - P_{\text{all}}H_{\text{all}} - E_0) \\ Y_2^* = Y_2 + \mu(H_{\text{all}} - H_{\text{all}}Z_{\text{all}} - E_2) \\ Y_3^{v*} = Y_3^v + \mu(X^v - P^vH^v - E_1^v) \\ Y_4^{v*} = Y_4^v + \mu(H^v - H^vZ^v - E_3^v) \\ Y_5^* = Y_5 + \mu(Z_{\text{all}} - J_{\text{all}}) \\ Y_6^{v*} = Y_6^v + \mu(Z^v - J^v + \text{Diag}(J^v)) \\ \mathcal{W}^* = \mathcal{W} + \oint(\mathcal{H} - \mathcal{Q}) \\ \mu = \min(\eta\mu, \mu_{\max}) \end{cases} \tag{29}$$

where η is a scaling factor, which enable μ to approach their maximum values $\mu_{\max}$. The convergence conditions of the algorithm are as follows:

$$\begin{cases} r_1 = \left\|X_{all} - P_{all}H_{all} - E_0\right\|_\infty \\ r_2 = \left\|H_{all} - H_{all}Z_{all} - E_2\right\|_\infty \\ r_3 = \left\|X^v - P^vH^v - E_1^v\right\|_\infty \\ r_4 = \left\|H^v - H^vZ^v - E_3^v\right\|_\infty \\ m_1 = \left\|Z_{all} - J_{all}\right\|_\infty \\ m_2 = \left\|Z^v - J^v + \text{Diag}(J^v)\right\|_\infty \\ m_3 = \left\|\mathcal{H} - \mathcal{Q}\right\|_\infty \\ \max\{r_1, r_2, r_3, r_4, m_1, m_2, m_3\} < \varepsilon \end{cases} \tag{30}$$

where ε is a predefined tolerance value. Algorithm 1 presents the optimization procedure of DSLSMSC in a step-by-step manner.

4.5. Complexity analysis

The time complexity of DSLSMSC mainly consists of solving ten subproblems: $Z_{all}, Z^v, P_{all}, P^v, H_{all}, H^v, E, \mathcal{Q}, J_{all}$, and J^v. The time complexities of solving the subproblem Z_{all} and Z^v are $O(n^3)$ and $O(vn^3)$ respectively. The time complexities of solving P_{all} and P^v are $O(k^2m + m^3)$ and $O(vn^3)$ respectively. When updating J_{all}, it involves SVD operation, and the related computational complexity is $O(n^3)$. The time complexity of solving J^v is $O(vn)$, which is caused by matrix subtraction. In the subproblems of updating H_{all} and H^v, the time complexities are $O(k^3)$ and $O(vn^3)$ respectively. The complexities of updating E_0, E_1^v, E_2 and E_3^v, which update each column in turn, are $O(mn^2)$, $O(vd_in^2)$, $O(mn^2)$ and $O(vd_in^2)$. When dealing with the $\mathcal{Q}$ subproblem, we need to perform fast Fourier transform, inverse fast Fourier transform and singular value decomposition on each frontal slice of the tensor. The computational complexity of this process is $O(vn^2\log(n))$. The total time complexity of DSLSMSC is $O(q(v(n^3 + d_in^2 + n + n^2log(n)) + n^3 + k^2m + m^3 + k^3 + mn^2))$, where q and v represent the number of iterations and the number of views respectively.

5. Experiment

We evaluate DSLSMSC through performance comparisons, clustering visualization, parameter sensitivity analysis, affinity matrix visualization, ablation studies, statistical tests, and convergence analysis. All experiments are implemented in MATLAB R2018b on a Windows 11 PC equipped with an Intel Core i9-13900H CPU @4.10 GHz and 16 GB RAM. Following the standard evaluation protocol in unsupervised multi-view clustering, all samples in each dataset are used to learn the affinity matrix and perform clustering, rather than being divided into training and test subsets.

5.1. Experimental setting

5.1.1. Datasets

We evaluate our method on eight widely used benchmark datasets: ORL, MSRC-V1, 100leaves, BBC, BBCSport, WebKB, Caltech101-7, and CiteSeer, with their detailed statistics summarized in Table 2. All datasets are normalized before being processed by DSLSMSC and the compared methods, reducing the influence of differences in feature scale.

ORL: The ORL dataset contains 400 grayscale face images, which are sourced from 40 separate individuals. Each person is depicted by 10 images that were taken at different points in time and under diverse lighting circumstances.

MSRC-V1: The MSRC-V1 dataset is a widely used multi-view object image dataset consisting of 210 samples from 7 categories. This dataset is commonly adopted to evaluate the performance of multi-view clustering algorithms.

100leaves: The 100leaves dataset is composed of leaf images representing 100 distinct types. For every leaf type, there are numerous sample images. These images are captured with different lighting setups and against diverse backgrounds. As a result, it forms a comprehensive

Algorithm 1: The optimization of DSLSMSC.

Input: Multi-view data $\{X^v\}$, parameters $\lambda_1, \lambda_2, \lambda_3, \mu, k$.

Initialized:

matrix:
$P_{\text{all}} = P^v = Z_{\text{all}} = Z^v = J_{\text{all}} = J^v = H_{\text{all}} = H^v = E_0 = E_1^v$
$= E_2 = E_3^v = Y_1$
$= Y_2 = Y_3^v = Y_4^v = Y_5 = Y_6^v = 0$
tensor:
$\mathcal{Q} = \mathcal{W} = \mathcal{H} = 0$
$\mu = 10^{-6}, \quad \eta = 2, \quad \varepsilon = 10^{-7}, \quad \mu_{\max} = 10^{10}$

while *not converged* **do**
  for $v = 1 : V$ **do**
    Update Z_{all} by Eq.(15) ;
    Update Z^v by Eq. (17);
    Update P_{all} by Eq.(18);
    Update P^v by Eq.(19);
    Update H_{all} by Eq.(21);
    Update H^v by Eq.(22);
    Update E by Eq.(23);
    Construct tensor $\mathcal{H} = \Phi(Z_{all}, Z^v)$;
    Update $\mathcal{Q}$ by Eq.(25);
    Update J_{all} by Eq.(26);
    Update J^v by Eq.(28);
    Update Y_1,Y_2,$\{Y_3^v\}$, $\{Y_4^v\}$, Y_5, $\{Y_6^v\}$, $\mathcal{W}$ and μ by Eq.(29);
  end
  $(\text{Q}\{1\}, \text{Q}\{v+1\}) = \Phi^{-1}(\mathcal{Q})$;
  Checking the convergence conditions.
end
Perform spectral clustering on
$\frac{1}{2}\left\{\left(\left|Z_{all}\right| + \left|Z_{all}{}^{\text{T}}\right|\right) + \sum_{v=1}^{V}\left(\left|Z^v\right| + \left|Z^{vT}\right|\right)\right\}$.
Output: Clustering results.

Table 2
Statistics of the benchmark datasets.

Datasets	Objective	Samples	Clusters	Dimensions
ORL	Face	400	40	[4096,3304,6750]
MSRC-V1	Object	210	7	[1302,48,512,100,256,210]
100leaves	Object	1600	100	[64,64,64]
BBCSport	Text	544	5	[3183,3203]
BBC	Text	685	6	[4659,4633,4665,4684]
WebKB	Web Page	1051	2	[2949,334]
Caltech101-7	Object	1474	7	[48,40,254,1984,512,928]
CiteSeer	Citation Network	3312	6	[3312,3703]

collection of images that is often utilized in research related to image classification and pattern recognition.

BBC: The BBC dataset contains 2225 news articles which are grouped into five separate classes. It is frequently used for text classification endeavors within the realm of natural language processing research.

BBCSport: The BBC dataset contains 685 news articles from six categories and is represented by four views.

WebKB: The dataset sourced from the websites of computer science departments at four universities encompasses web pages and focuses on two views, namely the text view on web pages and the hyperlink anchor text view.

Caltech101-7: The dataset is a subset of the Caltech101 dataset, containing 1474 samples from 7 object categories, including Faces, Motorbikes, dollar_bill, garfield, snoopy, stop_sign, and windsor_chair. Each category contains roughly 30 to 800 images. For each sample, six different views are provided, making this dataset widely used for evaluating multi-view clustering methods.

CiteSeer: The CiteSeer dataset is a citation network containing academic papers from six categories. Each paper is represented as a node with a feature vector and a class label, while citation relationships between papers are represented as edges.

5.1.2. Comparison methods

To evaluate the performance of DSLSMSC, we selected several state-of-the-art MVC methods for comparison, including the classical single-view LRR approach. Furthermore, DSLSMSC was comprehensively evaluated against multiple recently developed multi-view clustering techniques to demonstrate its effectiveness. To ensure optimal performance for all compared methods, we adopted the best parameter configurations reported in their respective original publications. To ensure fair statistical comparison, all experiments were conducted under an identical protocol and independently repeated 10 times. The reported mean and standard deviation were computed across these 10 runs. For some methods, zero standard deviation may still occur on certain datasets because the learned affinity matrices and the final clustering results are highly stable. In addition to reporting quantitative comparisons, we further provide a conceptual summary of several related dual-representation and tensor-based comparison methods in Table 3. This comparison helps clarify the methodological differences between DSLSMSC and recent baselines, especially in terms of representation design and structural constraints. Although these methods all aim to exploit consistency and complementarity across multiple views, they differ in how representations are constructed and how structural priors are imposed.

LRR_{best} (Liu et al., 2012): LRR is a classical single-view clustering method designed to identify low-dimensional subspace representations of data, effectively capturing global structural patterns. For multi-view data, we independently apply LRR to each individual view and select the optimal result among them.

CBFMSC (Zheng et al., 2020b):CBFMSC investigates inter-view relationships from a hierarchical perspective by introducing bilinear factorization into the coefficient matrix, which is theoretically well-founded.

FCMSC (Zheng et al., 2020a): FCMSC integrates multi-view data through feature concatenation to construct a unified feature representation. This concatenation-based approach preserves the original information from each individual view while enabling joint analysis.

t-SVD-MSC (Xie et al., 2018): t-SVD-MSC achieves robust and accurate multi-view clustering by representing multi-view data in tensor form and employing t-SVD to capture high-order correlations among multiple views.

LT-MSC (Zhang et al., 2015): LT-MSC leverages the high-order properties of tensors to overcome the limitation of conventional multi-view clustering methods that can only model pairwise relationships between views. By representing multi-view data as a tensor, this method simultaneously captures high-order correlations among multiple views, thereby enabling more comprehensive mining of structural information in the data.

Table 3
Conceptual comparison between DSLSMSC and closely related comparison methods.

Method	Representation level	Allocation of structural priors
DV-MSC (Lan et al., 2024)	Double-level view-correlation representation, including low-level feature-space representation and high-level subspace self-representation.	Uses HSIC-based diversity modeling at the low level and tensor low-rank constraints on stacked self-representation matrices at the high level.
TDCLM (Wu & Lu, 2025)	Graph-level consistency-diversity representation.	Decomposes multi-view relationships into diversity graphs and a consistency graph, then imposes tensor low-rank constraints and Laplacian manifold constraints on the constructed tensor.
TMSL (Cai et al., 2025)	Tensor multi-subspace representation built from the self-representation matrices generated by a typical TMVC method.	Stacks view-wise self-representation matrices into a tensor and introduces tensor low-rank representation to refine the tensor subspace structure. Sparse regularization is mainly used for modeling noise or error terms in the tensor representation process.
DSLSMSC	Latent representation-space decomposition, including one joint-view latent space and multiple original-view latent spaces.	Assigns low-rank regularization to the joint-view self-representation, sparse regularization to the original-view self-representations, and TNN regularization to cross-space fusion.

LMSC (Zhang et al., 2017): LMSC represents a novel latent multi-view subspace clustering approach that effectively uncovers latent complementary information by performing clustering through latent representations, thereby achieving more accurate subspace representations.

MCLES (Chen et al., 2020): The MCLES method achieves multi-view data clustering in a learned latent embedding space while simultaneously learning both global structures and cluster indicator matrices within a unified optimization framework.

DV-MSC (Lan et al., 2024): DV-MSC employs a novel framework that simultaneously performs latent mapping and subspace self-representation of multi-view data while incorporating a HSIC term to explore diversity in low-level feature representations. The method further stacks the self-representation matrices into a third-order tensor and applies low-rank tensor constraints to discover consistent high-level clustering structures.

TDCLM (Wu & Lu, 2025): TDCLM decomposes the self-representation matrix into a consensus graph and a diversity graph, which are then combined into a tensor structure with low-rank tensor constraints. The method further incorporates Laplacian manifold constraints to enhance the relationship between the diversity graph and consensus graph, thereby enabling more effective utilization of the intrinsic structural information in multi-view data.

PFMSCTGR (Ji & Feng, 2025): PFMSCTGR introduces a novel tensor rank algebra termed Tensor Graph Rank (TGR), which enforces a more constrained rank minimization through its unique regularization mechanism. Under TGR constraints, all non-zero singular values receive moderate-intensity penalization, a mathematical formulation that promotes enhanced discriminability in the resulting representation tensor.

SCDOD (Luo et al., 2025): SCDOD introduces slack consistency matrices to connect the self-representation matrices with the global consistency matrix, which helps capture global consistency information while reducing excessive consistency constraints. It further employs double-side orthogonal diversity constraints to enhance complementarity among different views by encouraging diversity between slack

Table 4
Experimental results on ORL (%).

Method	ACC	NMI	F-score	Precision	AR	Recall
LRR_{best}	76.88±2.01	88.55±1.17	69.33±2.69	65.66±2.63	68.58±2.76	73.46±2.92
CBFMSC	78.40±2.02	89.17±0.76	71.55±1.91	67.92±2.24	70.85±1.96	75.60±1.62
FCMSC	85.12±2.15	93.77±1.12	81.20±2.87	77.24±3.18	80.74±2.94	85.61±2.73
t-SVD-MSC	97.70±1.61	99.53±0.33	97.71±1.58	96.28±2.57	97.66±1.62	99.21±0.58
LT-MSC	81.63±2.98	91.93±1.42	76.52±3.43	72.10±3.80	75.95±3.52	81.56±3.29
LMSC	76.08±2.11	89.29±1.15	69.31±2.65	63.74±3.34	68.54±2.73	76.02±2.64
MCLES	82.15±1.04	91.12±0.49	73.90±1.46	68.71±2.13	73.25±1.50	79.98±1.13
DV-MSC	98.40±1.71	99.66±0.37	98.38±1.71	97.34±2.80	98.34±1.75	99.46±0.62
TDCLM	95.35±1.55	98.37±0.48	94.53±1.57	92.70±2.11	94.41±1.61	96.46±1.15
PFMSCTGR	83.95±1.30	92.76±0.91	78.74±2.14	74.56±2.59	78.23±2.20	83.46±2.17
SCDOD	96.28±1.00	98.87±0.32	95.51±1.05	93.66±1.52	95.51±1.05	97.65±0.76
TMSL	99.10±1.50	99.80±0.30	99.00±1.60	98.40±2.60	99.00±1.60	99.70±0.50
DSLSMSC	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**

Table 5
Experimental results on MSRC-V1 (%).

Method	ACC	NMI	F-score	Precision	AR	Recall
LRR_{best}	63.67±0.32	57.96±0.28	51.70±0.15	48.82±0.00	43.39±0.17	54.95±0.22
CBFMSC	91.29±0.23	83.64±0.51	83.48±0.42	82.76±0.38	80.79±0.49	84.21±0.46
FCMSC	70.95±0.45	57.86±0.52	56.16±0.56	55.76±0.54	49.04±0.65	56.57±0.58
t-SVD-MSC	99.52±0.00	98.92±0.00	99.03±0.00	99.02±0.00	98.88±0.00	99.05±0.00
LT-MSC	84.38±0.30	75.56±0.39	72.50±0.58	70.87±0.74	67.95±0.69	74.21±0.41
LMSC	82.48±0.30	72.09±0.42	70.46±0.47	69.77±0.59	65.64±0.55	71.16±0.37
MCLES	89.95±0.15	81.69±0.32	80.78±0.23	79.76±0.24	77.64±0.27	81.83±0.23
DV-MSC	99.52±0.00	98.92±0.00	99.03±0.00	99.02±0.00	98.88±0.00	99.05±0.00
TDCLM	98.57±0.00	96.77±0.00	97.10±0.00	97.05±0.00	96.63±0.00	97.14±0.00
PFMSCTGR	99.05±0.00	97.85±0.00	97.75±0.00	98.03±0.00	97.75±0.00	98.10±0.00
SCDOD	98.10±0.00	95.70±0.00	96.18±0.00	96.13±0.00	95.56±0.00	96.22±0.00
TMSL	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
DSLSMSC	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**

Table 6
Experimental results on 100leaves (%).

Method	ACC	NMI	F-score	Precision	AR	Recall
LRR_{best}	49.09±0.99	72.45±0.31	36.33±0.91	34.79±0.98	35.70±0.92	38.00±0.88
CBFMSC	78.98±1.48	90.24±0.74	71.80±1.77	68.40±1.75	71.52±1.79	75.56±1.87
FCMSC	76.14±1.10	88.51±0.52	67.59±1.34	64.18±1.50	67.27±1.36	71.39±1.31
t-SVD-MSC	92.49±1.01	98.50±0.28	92.19±1.66	88.92±1.64	92.11±1.68	96.93±0.49
LT-MSC	73.82±2.54	87.18±1.24	64.68±2.98	61.56±3.16	64.32±3.02	68.13±2.78
LMSC	75.38±0.99	88.16±0.55	66.58±1.10	63.29±1.01	66.24±1.11	70.24±1.57
MCLES	51.61±2.22	79.64±0.84	41.16±3.13	30.13±3.10	40.39±3.19	65.35±1.14
DV-MSC	92.22±1.03	98.14±0.26	91.84±1.06	88.36±1.48	91.76±1.07	95.61±0.71
TDCLM	89.02±1.50	97.33±0.33	88.20±1.62	83.42±2.41	88.08±1.63	93.59±0.80
PFMSCTGR	92.14±1.22	98.35±0.29	92.00±1.25	87.83±1.77	91.93±1.27	96.60±0.77
SCDOD	90.93±1.20	97.92±0.31	90.53±1.20	86.45±1.73	90.43±1.21	95.02±0.79
TMSL	92.30±0.70	98.30±0.10	91.40±1.00	86.70±1.80	91.30±1.10	96.70±0.20
DSLSMSC	**92.77±1.97**	**98.67±0.36**	**93.03±1.83**	**89.26±2.74**	**92.96±1.85**	**97.16±0.84**

consistency matrices and specificity matrices, as well as among the specificity matrices themselves.

TMSL (Cai et al., 2025): TMSL first employs a typical tensor-based multi-view clustering method to obtain self-representation matrices for each view, and then introduces tensor low-rank representation to uncover the intrinsic tensor subspace structure of the data. By integrating these two stages into a unified framework, TMSL learns a reliable and robust multi-subspace representation tensor.

5.1.3. Evaluation metric

Consistent with the benchmark MVC method (Xie et al., 2018), we employ six quantitative metrics to evaluate model performance: Accuracy (ACC), Normalized Mutual Information (NMI), F-Score, Precision, Recall, and Adjusted Rand Index (ARI), where higher values across all metrics indicate better clustering performance.

5.2. Experimental results and analysis

5.2.1. Performance comparison

Tables 4–11 report the mean and standard deviation of DSLSMSC and the compared methods on the benchmark datasets over 10 runs, where the best results are highlighted in bold and the second-best results are underlined. Several observations can be made from these results. The single-view baseline LRR_{best} consistently performs worse than most multi-view methods, indicating that relying on only one view is generally insufficient to capture the intrinsic structure of heterogeneous data. Compared with LMSC, DSLSMSC generally obtains higher scores on the evaluated datasets, suggesting that combining information from the joint view and original view spaces is beneficial for clustering. On ORL and MSRC-V1, DSLSMSC achieves saturated performance, reaching 100.00% on all evaluation metrics; this should be understood as

Table 7
Experimental results on BBCSport (%).

Method	ACC	NMI	F-score	Precision	AR	Recall
LRR_{best}	76.25±1.34	61.63±0.89	70.17±0.85	71.72±1.51	61.08±1.23	68.69±0.24
CBFMSC	90.26±0.00	82.27±0.00	88.86±0.00	88.45±0.00	85.35±0.00	89.28±0.00
FCMSC	87.87±0.00	69.82±0.00	80.46±0.00	81.10±0.00	74.40±0.00	79.84±0.00
t-SVD-MSC	97.06±0.00	90.49±0.00	93.71±0.00	93.13±0.00	91.72±0.00	94.29±0.00
LT-MSC	52.22±7.85	27.13±10.18	46.65±5.29	33.19±6.10	18.87±11.08	81.80±6.27
LMSC	91.54±0.00	83.87±0.00	89.87±0.00	89.20±0.00	86.66±0.00	90.55±0.00
MCLES	96.14±0.00	87.66±0.00	92.29±0.00	91.44±0.00	89.85±0.00	93.16±0.00
DV-MSC	98.90±0.00	96.29±0.00	98.13±0.00	98.57±0.00	97.54±0.00	97.69±0.00
TDCLM	96.14±0.00	90.89±0.00	94.93±0.00	94.97±0.00	93.34±0.00	94.88±0.00
PFMSCTGR	94.12±0.00	83.04±0.00	88.84±0.00	88.12±0.00	85.30±0.00	89.57±0.00
SCDOD	99.26±0.00	97.30±0.00	98.78±0.00	99.11±0.00	98.40±0.00	98.45±0.00
TMSL	99.60±0.00	98.70±0.00	99.10±0.00	99.30±0.00	98.80±0.00	98.90±0.00
DSLSMSC	**99.82±0.00**	**99.30±0.00**	**99.68±0.00**	**99.65±0.00**	**99.58±0.00**	**99.71±0.00**

Table 8
Experimental results on BBC (%).

Method	ACC	NMI	F-score	Precision	AR	Recall
LRR_{best}	57.01±0.62	36.40±0.16	47.95±0.62	51.61±0.57	33.45±0.76	44.78±0.66
CBFMSC	84.38±0.00	68.77±0.00	74.61±0.00	77.35±0.00	67.18±0.00	72.05±0.00
FCMSC	60.28±5.86	45.09±1.21	53.34±4.47	54.34±2.20	39.47±5.01	52.55±6.60
t-SVD-MSC	95.18±0.00	90.84±0.00	91.46±0.00	94.82±0.00	88.96±0.00	88.32±0.00
LT-MSC	91.11±0.00	75.71±0.16	84.05±0.00	83.82±0.00	79.15±0.00	84.29±0.00
LMSC	83.26±0.00	60.44±0.00	73.45±0.00	73.86±0.00	65.38±0.11	73.05±0.11
MCLES	84.00±0.00	68.39±0.28	78.26±0.12	79.01±0.13	71.69±0.16	77.53±0.14
DV-MSC	96.50±0.00	88.58±0.00	93.40±0.00	94.06±0.00	91.40±0.00	92.75±0.00
TDCLM	95.18±0.00	90.27±0.00	93.13±0.00	95.19±0.00	91.09±0.00	91.17±0.00
PFMSCTGR	92.41±0.00	78.49±0.00	86.59±0.00	86.31±0.00	82.47±0.00	86.87±0.00
SCDOD	98.54±0.00	94.96±0.00	97.39±0.00	97.42±0.00	96.59±0.00	97.36±0.00
TMSL	97.50±0.00	93.40±0.00	95.40±0.00	97.40±0.00	94.10±0.00	93.60±0.00
DSLSMSC	**99.12±0.00**	**97.30±0.00**	**98.44±0.00**	**99.14±0.00**	**97.97±0.00**	**97.76±0.00**

Table 9
Experimental results on WebKB (%).

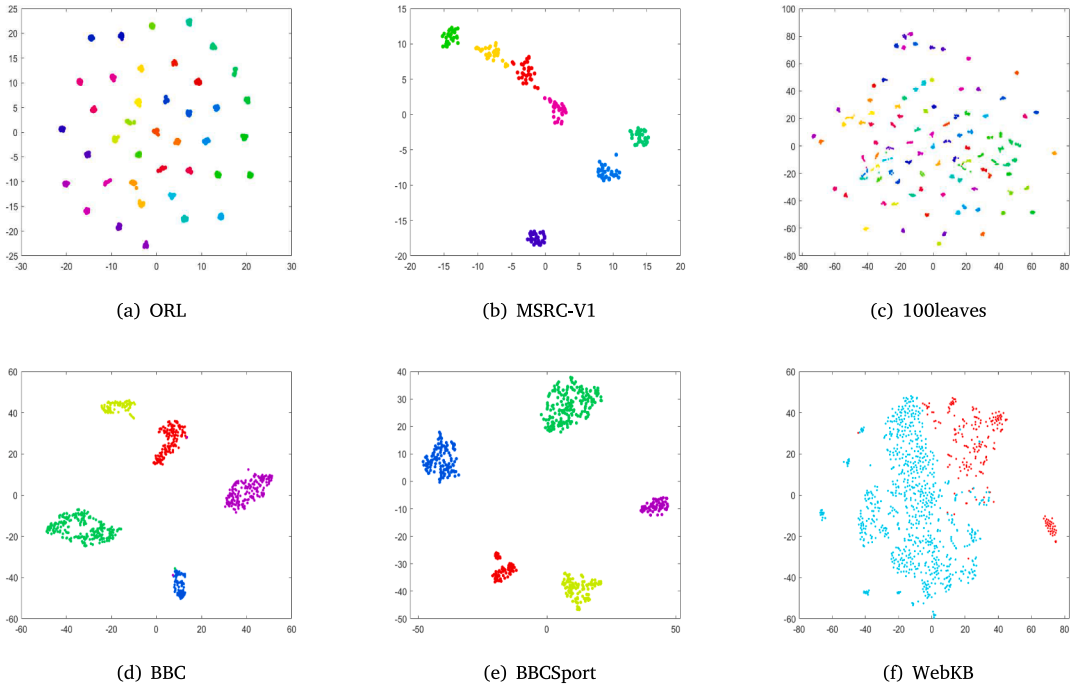
Method	ACC	NMI	F-score	Precision	AR	Recall
LRR_{best}	77.41±0.00	0.23±0.00	78.76±0.00	65.99±0.00	1.15±0.00	97.68±0.00
CBFMSC	82.70±0.00	19.38±0.00	81.75±0.00	70.78±0.00	24.05±0.00	96.74±0.00
FCMSC	84.02±0.00	38.48±0.00	77.58±0.00	85.94±0.00	44.80±0.00	70.70±0.00
t-SVD-MSC	96.86±0.00	79.21±0.00	95.25±0.00	97.80±0.00	86.79±0.00	92.84±0.00
LT-MSC	76.16±0.00	5.70±0.00	78.44±0.00	68.74±0.00	13.56±0.00	91.33±0.00
LMSC	80.97±0.00	10.68±0.00	79.62±0.00	70.40±0.00	20.43±0.00	91.62±0.00
MCLES	86.20±0.00	31.51±0.00	81.49±0.00	83.40±0.00	48.19±0.00	79.66±0.00
DV-MSC	90.58±0.00	43.71±0.00	87.53±0.00	84.19±0.00	60.56±0.00	91.15±0.00
TDCLM	82.02±0.00	14.18±0.00	80.76±0.00	70.66±0.00	22.50±0.00	94.24±0.00
PFMSCTGR	77.35±0.00	0.06±0.00	78.41±0.00	65.89±0.00	0.62±0.00	96.80±0.00
SCDOD	96.86±0.00	76.63±0.00	95.29±0.00	97.07±0.00	86.70±0.00	93.57±0.00
TMSL	97.40±0.00	78.50±0.00	96.20±0.00	96.70±0.00	88.90±0.00	95.60±0.00
DSLSMSC	**97.72±0.00**	**83.42±0.00**	**96.54±0.00**	**98.42±0.00**	**90.25±0.00**	**94.73±0.00**

Table 10
Experimental results on Caltech101-7 (%).

Method	ACC	NMI	F-score	Precision	AR	Recall
LRR_{best}	40.62±0.18	29.03±0.11	39.07±0.05	67.25±0.09	21.49±0.06	27.54±0.04
CBFMSC	56.23±0.02	56.90±0.22	54.51±0.05	81.73±0.04	38.77±0.04	40.89±0.06
FCMSC	60.74±0.03	58.53±0.03	60.88±0.03	86.15±0.01	46.24±0.03	47.08±0.04
t-SVD-MSC	64.84±4.64	73.36±2.75	68.28±3.67	96.53±0.97	56.46±4.22	52.89±3.98
LT-MSC	60.87±0.02	63.47±0.01	61.54±0.02	89.84±0.01	47.74±0.02	46.80±0.02
LMSC	55.47±0.06	50.66±0.04	54.27±0.04	82.65±0.06	38.77±0.05	40.39±0.03
MCLES	59.70±0.00	58.39±0.04	59.74±0.01	88.44±0.02	45.57±0.02	45.11±0.03
DV-MSC	64.31±0.02	68.11±0.05	67.43±0.02	93.68±0.05	54.92±0.03	52.67±0.03
TDCLM	43.22±0.15	53.79±0.12	47.24±0.05	84.66±0.06	32.78±0.05	32.76±0.04
PFMSCTGR	65.96±0.02	75.15±0.02	68.45±0.02	97.04±0.01	56.50±0.02	53.04±0.02
SCDOD	66.88±0.16	67.49±0.10	68.39±0.11	93.87±0.33	56.05±0.13	52.95±0.19
TMSL	63.70±0.50	66.40±0.20	51.00±0.02	92.70±0.20	51.00±0.06	49.00±0.60
DSLSMSC	**67.35±0.07**	**75.49±0.17**	**68.67±0.09**	**97.09±0.03**	**56.92±0.11**	**53.12±0.09**

Table 11
Experimental results on CiteSeer (%).

Method	ACC	NMI	F-score	Precision	AR	Recall
LRR_{best}	44.37±0.05	20.64±0.37	33.66±0.05	27.70±0.05	15.28±0.07	42.88±0.04
CBFMSC	57.55±0.00	30.59±0.00	41.61±0.00	42.15±0.00	29.12±0.00	41.09±0.00
FCMSC	47.66±0.01	23.60±0.01	34.98±0.01	34.65±0.01	20.68±0.01	35.30±0.01
t-SVD-MSC	57.29±0.01	33.20±0.01	41.99±0.01	41.19±0.01	29.08±0.02	42.82±0.01
LT-MSC	60.39±0.01	32.64±0.01	44.32±0.01	45.09±0.01	32.47±0.01	43.59±0.01
LMSC	56.11±0.03	29.83±0.04	40.36±0.03	40.61±0.04	27.50±0.04	40.11±0.03
MCLES	42.18±0.00	17.22±0.00	30.01±0.00	30.79±0.00	15.26±0.00	29.27±0.00
DV-MSC	45.01±0.02	20.44±0.01	31.91±0.01	32.56±0.01	17.47±0.02	31.28±0.02
TDCLM	55.10±0.00	29.28±0.00	39.74±0.00	39.95±0.00	26.73±0.00	39.53±0.00
PFMSCTGR	58.35±0.02	34.31±0.02	43.36±0.01	42.63±0.02	30.79±0.02	44.11±0.01
SCDOD	53.58±0.06	27.28±0.05	37.22±0.06	37.49±0.06	23.69±0.07	36.95±0.05
TMSL	58.00±0.00	35.60±0.00	44.80±0.00	44.30±0.00	32.70±0.00	45.40±0.00
DSLSMSC	**61.85±0.02**	**35.73±0.03**	**45.64±0.02**	**45.55±0.02**	**33.80±0.02**	**45.73±0.02**



(a) ORL (b) MSRC-V1 (c) 100leaves

(d) BBC (e) BBCSport (f) WebKB

Fig. 2. t-SNE of the DSLSMSC.

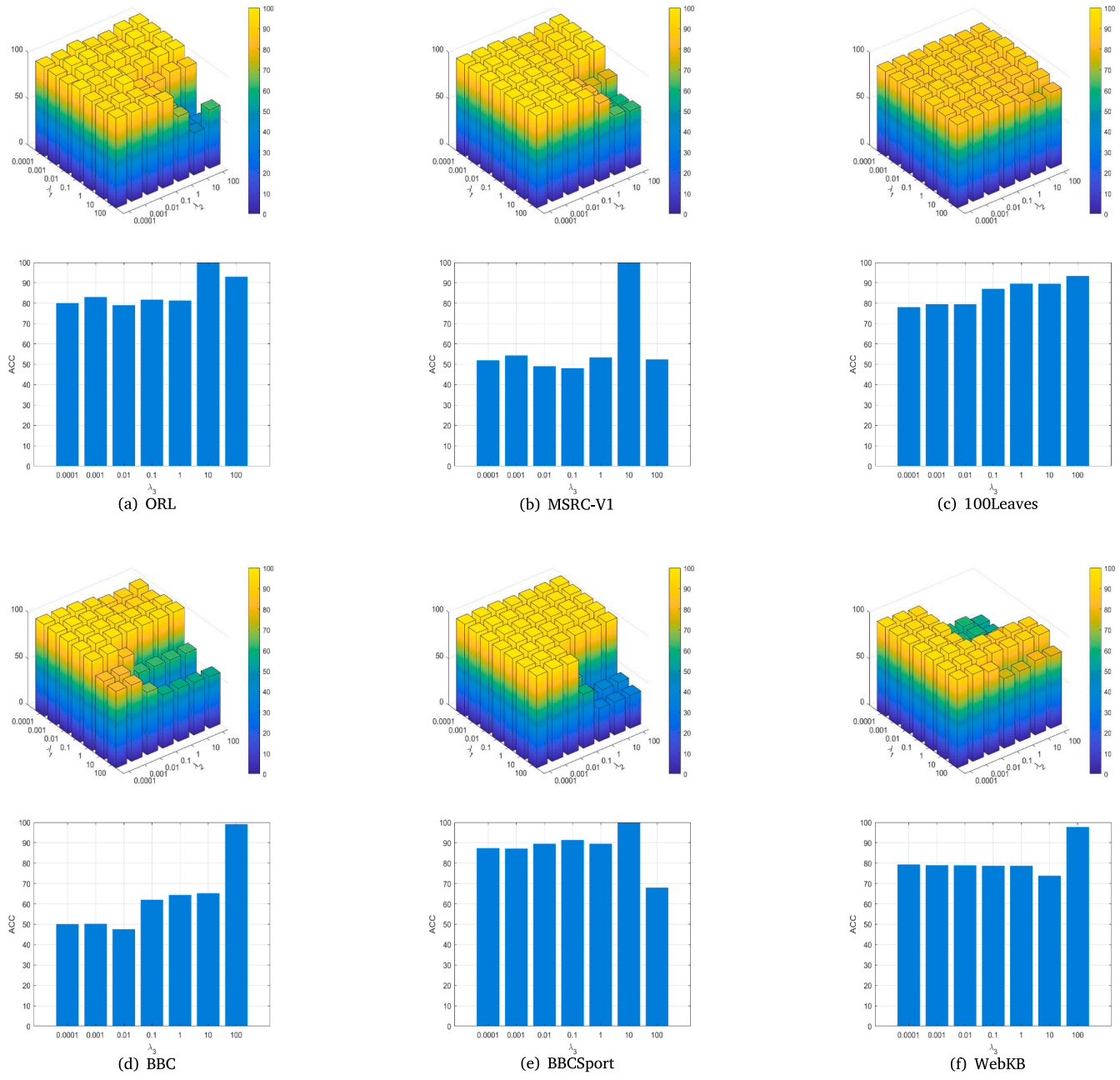
benchmark saturation on relatively small datasets with clear cluster structures. On the larger or more challenging datasets, DSLSMSC also performs competitively. On 100leaves, BBCSport, BBC, WebKB, Caltech101-7 and CiteSeer, it achieves the best ACC and NMI among the compared methods, and it remains competitive on the other evaluation metrics as well. For example, on BBC, DSLSMSC improves ACC and NMI from 98.54% and 94.96% to 99.12% and 97.30%, respectively, compared with the best performing baseline on these two metrics.

5.2.2. Results visualization

We conducted comprehensive visualization of affinity matrix structural representations using t-distributed stochastic neighbor embedding (t-SNE) on six distinct datasets, where Fig. 2 demonstrates that DSLSMSC generates exceptionally compact cluster structures (with distinct colors representing different clusters), visually confirming its superior capability in excavating intrinsic data structural information for enhanced clustering performance.

5.2.3. Parameter setting

In this subsection, we analyze three parameters $\lambda_1, \lambda_2, \lambda_3$ and the dimensionality k of the latent space. Specifically, the three parameters are adjusted within the range [0.0001, 0.001, 0.01, 0.1, 1, 10, 100]. Through this range, we explore the optimal parameters for all datasets mentioned in this paper. Fig. 3 presents the parameter sensitivity. We choose to investigate the impact of the other two parameters on ACC by fixing one of them. The adjustment of λ_1 and λ_2 has a minor effect on the 100leaves dataset, while it has a certain influence on the other datasets. However, the overall impact remains relatively stable, indicating that DSLSMSC possesses a certain degree of robustness. For λ_3, precise tuning is particularly crucial. According to our experiments, the parameter λ_3 is recommended to be selected from the range $[1, 100]$, since it controls the strength of cross-space fusion and has a relatively direct influence on clustering performance. Additionally, we fix the dimensionality of the latent space at 100, i.e., $k = 100$.The selection of k follows the principle that the parameter k is selected from the interval $[c, n]$, where c represents the number of clusters and n denotes the total number of



(a) ORL (b) MSRC-V1 (c) 100Leaves

(d) BBC (e) BBCSport (f) WebKB

Fig. 3. Performance of DSLSMSC with varying parameters λ_1 and λ_2.

samples in the dataset. This selection criterion ensures that k maintains an appropriate intermediate value between these two critical bounds. To further illustrate the influence of k, we take the MSRC-V1 dataset as an example and vary k within the interval $[10, 210]$. The experimental results are shown in Fig. 4. This demonstrates that the choice of dimensionality k is crucial for clustering performance. An excessively small k prevents the projection matrix from adequately capturing the intrinsic structural information of the data, leading to the loss of significant key features during projection and resulting in overly coarse clustering outcomes, thereby reducing clustering accuracy and effectiveness. Conversely, an excessively large k introduces excessive noise and redundant information, making the projection matrix complex and unstable. For small- to medium-scale datasets, we recommend selecting k from the range $[50, 150]$.

5.2.4. *Running time*

Table 12 reports the running time (in seconds) on seven benchmark datasets. Overall, DSLSMSC generally requires more computation time than some lightweight baselines because it jointly optimizes representations in two spaces and further performs tensor regularization. Nevertheless, its running time remains competitive compared with several recent complex baselines. It should be noted that the current DSLSMSC model is mainly suitable for datasets of small or medium size. Owing to the cubic time complexity scaling with sample numbers, as well as iterative SVD and t-SVD computations, the current implementation is not suitable for ultra-large-scale datasets such as those with $N > 10{,}000$. Therefore, this should be regarded as a practical boundary condition of the present method.

Table 12
Running times (in seconds) required on the benchmark datasets.

Methods	ORL	MSRC-V1	100leaves	BBCSport	BBC	WebKB	Caltech101-7	CiteSeer
LRR_{best}	27.6	1.5	106.7	14.6	45.1	82.6	92.1	4215.4
CBFMSC	34.9	3.3	63.9	21.2	76.1	38.8	149.5	901.2
FCMSC	13.5	3.9	98.7	13.5	48.1	38.7	203.9	1321.8
t-SVD-MSC	23.7	3.2	197.6	22.7	354.8	26.1	141.1	681.4
LT-MSC	44.5	9.5	215.0	28.9	133.1	76.4	427.2	2484.8
LMSC	32.1	5.1	138.5	40.6	88.7	60.4	119.9	1427.9
MCLES	50.9	3.2	9671.4	105.7	256.1	1295.1	9429.9	223114.8
DV-MSC	199.9	19.5	835.6	84.9	319.9	437.6	1479.4	4691.8
TDCLM	18.3	2.2	112.9	18.1	49.2	72.9	167.0	3402.9
PFMSCTGR	106.3	13.8	586.5	38.4	88.7	264.4	642.5	4739.8
SCDOD	16.0	5.1	66.5	11.7	84.2	31.1	214.5	742.5
TMSL	31.1	13.1	358.6	25.8	97.92	66.6	537.8	1068.2
DSLSMSC	40.5	12.1	569.4	36.7	146.5	106.8	526.5	5607.3

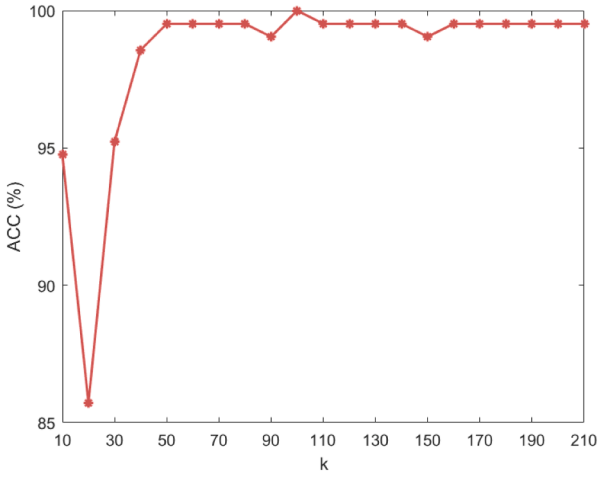


Fig. 4. Parameter tuning of k.

5.2.5. Ablation study

To systematically evaluate the effectiveness of individual components in DSLSMSC, we conducted comprehensive ablation studies across seven benchmark datasets. Specifically, we created three variants: (1)DSLSMSC-a by removing the joint-space learning module, (2)DSLSMSC-b (equivalent to LMSC) by eliminating the original-space learning component, and (3)DSLSMSC-c by discarding the TNN constraint. As quantitatively demonstrated in Table 13, DSLSMSC consistently outperforms all variants (DSLSMSC-a/b/c) across all testing scenarios, with DSLSMSC-c showing particularly degraded performance due to its inability to effectively fuse cross-space feature matrices without TNN regularization. These findings conclusively validate the necessity of our full framework, where the dual-space strategy simultaneously learns global and local structures in latent spaces while the TNN constraint enables optimal feature matrix fusion, collectively achieving significant performance gains in multi-view data clustering.

To further validate the effectiveness of the proposed modules, especially the rationality of the dual-space constraint allocation strategy, we additionally designed three variants. In the first variant, low-rank regularization is imposed on both the joint-view space and the original-view spaces, which is denoted as DSLSMSC-L/L. In the second variant, sparse regularization is imposed on both spaces, which is denoted as DSLSMSC-S/S. In the third variant, denoted as DSLSMSC-Swap, the low-rank and sparse regularizations are exchanged across the two spaces, that is, sparse regularization is used in the joint-view space while low-rank regularization is used in the original-view spaces. The corresponding experimental results are reported in Table 14. The full DSLSMSC achieves the best overall performance across all benchmark datasets, indicating that the effectiveness of the proposed method does not arise from simply introducing additional regularizers, but from assigning different regularizations to different spaces according to their structural roles. Applying the same constraint to both spaces is consistently less effective than the proposed design. For example, on the WebKB dataset, the best same-constraint variant, DSLSMSC-S/S, achieves an ACC of 94.96%, whereas DSLSMSC reaches 97.72%. On the Caltech101-7 dataset, the best same-constraint variant is DSLSMSC-L/L, which achieves an ACC of 66.24%, whereas DSLSMSC achieves 67.35%. In addition, exchanging the two constraints across spaces, as in DSLSMSC-Swap, also leads to performance degradation on most datasets. These results suggest that the joint-view space is more suitable for low-rank modeling of shared global structure, whereas the original-view spaces are more suitable for sparse modeling of view-specific local structure. Therefore, the additional ablation results provide stronger empirical support for the necessity of the proposed module design and the effectiveness of the dual-space constraint allocation strategy.

5.3. Statistical analysis

The Friedman test basis:

$$\mathrm{T}_F = \frac{(Q-1)T_{\chi^2}}{Q(A-1)-T_{\chi^2}} \tag{31}$$

where Q is the number of datasets and A is the number of methods.

The Friedman statistic is defined as

$$T_{\chi^2} = \frac{12Q}{A(A+1)}\left(\sum_{i=1}^{A} r_i^2 - \frac{A(A+1)^2}{4}\right), \tag{32}$$

where r_i represents the average ranking value of the ith algorithm. If T_F exceeds the critical value corresponding to the Friedman test, then continue with the Nemenyi test.

The Nemenyi test basis:

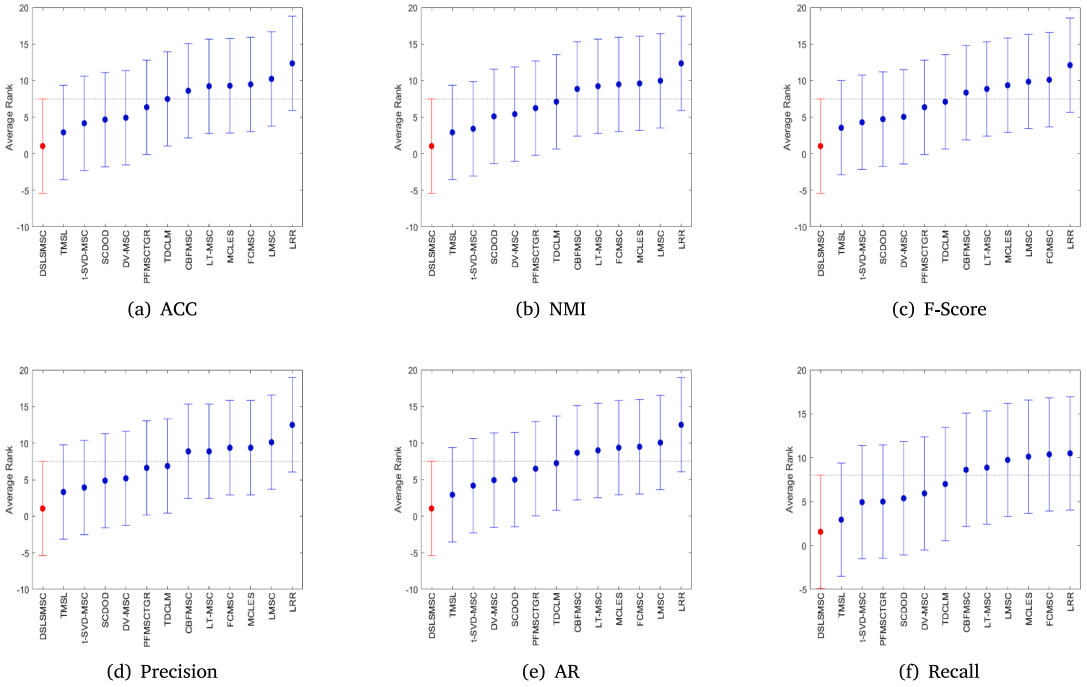
$$CD = q_\alpha \sqrt{\frac{A(A+1)}{6Q}} \tag{33}$$

q_α is the critical value of the Tukey distribution and CD is the critical value of the Nemenyi test. If the difference between the average order values of the two algorithms exceeds the critical value range CD, it can be considered that there is a difference in performance between the two algorithms; otherwise, there is no difference.

In our experiment, $Q = 8$ and $A = 13$. With $\alpha = 0.05$, the T_F values for ACC, NMI, F-score, Precision, AR, and Recall are reported in Table 15 based on Eq. (31). All these values are larger than the critical value 1.8693, indicating that the null hypothesis that all algorithms achieve equivalent performance should be rejected. Therefore, we further conduct the Nemenyi post hoc test. According to Eq. (33), the critical difference is $CD = 6.4505$. For a more intuitive comparison between DSLSMSC and the competing methods, the corresponding critical difference diagrams are presented in Fig. 5. As shown, DSLSMSC achieves the best

Table 13
Ablation study: Comparison results produced by DSLSMSC(%).

Dataset	Method	ACC	NMI	F-Score	Precision	AR	Recall
ORL	DSLSMSC-a	98.73±1.65	99.73±0.34	98.70±1.67	97.87±2.75	98.67±1.71	99.57±0.57
	DSLSMSC-b	76.08±2.11	89.29±1.15	69.31±2.65	63.74±3.34	68.54±2.73	76.02±2.64
	DSLSMSC-c	82.60±2.11	92.58±0.86	76.83±2.34	71.62±2.68	76.26±2.40	82.89±2.09
	DSLSMSC	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
MSRC-V1	DSLSMSC-a	99.52±0.00	98.92±0.00	99.03±0.00	99.02±0.00	98.88±0.00	99.05±0.00
	DSLSMSC-b	82.48±0.30	72.09±0.42	70.46±0.47	69.77±0.59	65.64±0.55	71.16±0.37
	DSLSMSC-c	90.10±0.20	80.88±0.23	81.43±0.30	80.73±0.29	78.41±0.35	82.13±0.32
	DSLSMSC	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
100leaves	DSLSMSC-a	90.66±1.20	97.11±0.53	89.31±1.61	85.74±1.91	89.20±1.62	93.19±1.32
	DSLSMSC-b	73.82±2.54	87.18±1.24	64.68±2.98	61.56±3.16	64.32±3.02	68.13±2.78
	DSLSMSC-c	80.16±1.92	90.97±0.64	73.56±2.02	70.09±2.30	73.30±2.04	77.41±1.78
	DSLSMSC	**92.77±1.97**	**98.67±0.36**	**93.03±1.83**	**89.26±2.74**	**92.96±1.85**	**97.16±0.84**
BBCSport	DSLSMSC-a	98.90±0.00	96.28±0.00	98.16±0.00	98.57±0.00	97.58±0.00	97.75±0.002
	DSLSMSC-b	91.54±0.00	83.87±0.00	89.87±0.00	89.20±0.00	86.66±0.00	90.55±0.00
	DSLSMSC-c	76.29±0.00	65.49±0.00	73.63±0.00	69.78±0.00	64.76±0.0	77.93±0.00
	DSLSMSC	**99.82±0.00**	**99.30±0.00**	**99.68±0.00**	**99.65±0.00**	**99.58±0.00**	**99.71±0.00**
BBC	DSLSMSC-a	98.10±0.00	93.87±0.00	96.71±0.00	97.50±0.00	95.71±0.00	95.93±0.00
	DSLSMSC-b	83.26±0.00	60.44±0.00	73.45±0.00	73.86±0.00	65.38±0.11	73.05±0.11
	DSLSMSC-c	82.42±0.26	58.89±0.31	77.28±0.31	72.46±0.20	63.82±0.39	72.11±0.43
	DSLSMSC	**99.12±0.00**	**97.30±0.00**	**98.44±0.00**	**99.14±0.00**	**97.97±0.00**	**97.76±0.00**
WebKB	DSLSMSC-a	95.00±0.00	71.55±0.18	92.47±0.00	96.38±0.00	79.59±0.18	88.86±0.10
	DSLSMSC-b	80.97±0.00	10.68±0.00	79.62±0.00	70.40±0.00	20.43±0.00	91.62±0.001
	DSLSMSC-c	80.78±0.00	15.24±0.00	76.65±0.00	75.77±0.00	30.21±0.00	77.54±0.00
	DSLSMSC	**97.72±0.00**	**83.42±0.00**	**96.54±0.00**	**98.42±0.00**	**90.25±0.00**	**94.73±0.00**
Caltech101-7	DSLSMSC-a	65.20±0.00	75.17±0.05	67.57±0.01	96.41±0.00	55.73±0.01	51.72±0.01
	DSLSMSC-b	55.47±0.06	50.66±0.02	54.27±0.04	82.65±0.06	38.77±0.05	40.39±0.03
	DSLSMSC-c	61.56±0.06	63.25±0.02	62.05±0.06	88.94±0.02	48.09±0.07	47.64±0.07
	DSLSMSC	**67.35±0.07**	**75.49±0.17**	**68.67±0.09**	**97.09±0.03**	**56.92±0.11**	**53.12±0.09**
CiteSeer	DSLSMSC-a	56.78±0.04	29.19±0.04	40.22±0.03	41.02±0.03	27.54±0.04	39.45±0.03
	DSLSMSC-b	56.11±0.03	29.83±0.04	40.36±0.03	40.61±0.04	27.50±0.04	40.11±0.03
	DSLSMSC-c	59.24±0.03	32.67±0.03	43.61±0.03	43.61±0.03	31.36±0.03	43.62±0.03
	DSLSMSC	**61.85±0.02**	**35.73±0.03**	**45.64±0.02**	**45.55±0.02**	**33.80±0.02**	**45.73±0.02**



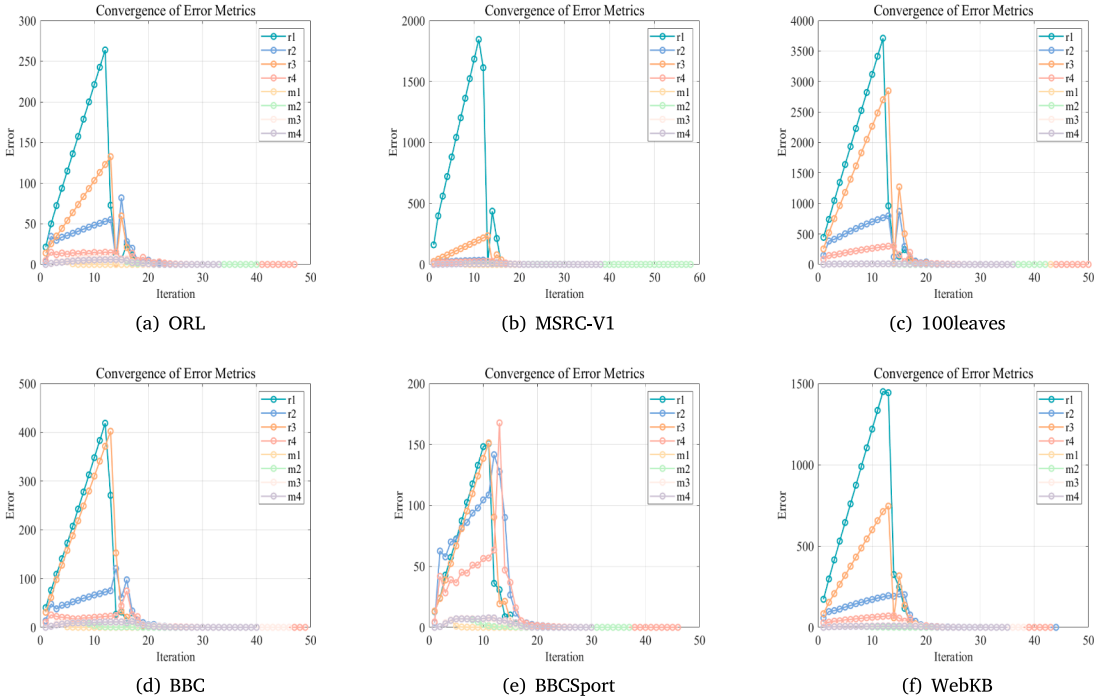
(a) ACC (b) NMI (c) F-Score

(d) Precision (e) AR (f) Recall

Fig. 5. Friedman test on six evaluation indicators.

Table 14
Additional ablation study on different constraint allocation strategies (%).

Dataset	Method	ACC	NMI	F-Score	Precision	AR	Recall
ORL	DSLSMSC-L/L	98.20±1.90	99.65±0.37	98.31±1.78	97.33±2.81	98.27±1.82	99.33±0.71
	DSLSMSC-S/S	99.08±1.51	99.81±0.30	99.07±1.50	98.41±2.56	99.05±1.54	99.76±0.45
	DSLSMSC-Swap	99.08±1.51	99.81±0.30	99.07±1.50	98.41±2.56	99.05±1.54	99.76±0.45
	DSLSMSC	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
MSRC-V1	DSLSMSC-L/L	99.52±0.00	98.92±0.00	99.03±0.00	99.02±0.00	98.88±0.00	99.05±0.00
	DSLSMSC-S/S	99.52±0.00	98.92±0.00	99.03±0.00	99.02±0.00	98.88±0.00	99.05±0.00
	DSLSMSC-Swap	99.52±0.00	98.92±0.00	99.03±0.00	99.02±0.00	98.88±0.00	99.05±0.00
	DSLSMSC	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
100leaves	DSLSMSC-L/L	91.92±0.99	98.35±0.28	91.87±1.24	87.89±1.75	91.79±1.26	96.23±0.88
	DSLSMSC-S/S	90.59±0.97	97.97±0.41	90.37±1.32	85.96±1.53	90.28±1.33	95.27±1.10
	DSLSMSC-Swap	91.38±1.34	98.12±0.35	91.11±1.56	87.08±2.26	91.02±1.57	95.55±0.94
	DSLSMSC	**92.77±1.97**	**98.67±0.36**	**93.03±1.83**	**89.26±2.74**	**92.96±1.85**	**97.16±0.84**
BBCSport	DSLSMSC-L/L	99.45±0.00	98.00±0.00	98.78±0.00	98.94±0.00	98.40±0.00	98.62±0.00
	DSLSMSC-S/S	99.45±0.00	98.00±0.00	98.78±0.00	98.94±0.00	98.40±0.00	98.62±0.00
	DSLSMSC-Swap	99.45±0.00	98.00±0.00	98.78±0.00	98.94±0.00	98.40±0.00	98.62±0.00
	DSLSMSC	**99.82±0.00**	**99.30±0.00**	**99.68±0.00**	**99.65±0.00**	**99.58±0.00**	**99.71±0.00**
BBC	DSLSMSC-L/L	98.83±0.00	96.67±0.00	97.89±0.00	98.83±0.00	97.26±0.00	96.98±0.00
	DSLSMSC-S/S	98.54±0.00	95.79±0.00	97.29±0.00	98.42±0.00	96.48±0.00	96.19±0.00
	DSLSMSC-Swap	98.54±0.00	95.79±0.00	97.29±0.00	98.42±0.00	96.48±0.00	96.19±0.00
	DSLSMSC	**99.12±0.00**	**97.30±0.00**	**98.44±0.00**	**99.14±0.00**	**97.97±0.00**	**97.76±0.00**
WebKB	DSLSMSC-L/L	93.34±0.00	59.62±0.00	90.23±0.00	93.29±0.00	73.14±0.00	87.36±0.00
	DSLSMSC-S/S	94.96±0.00	71.41±0.00	92.42±0.00	96.36±0.00	79.45±0.00	88.79±0.00
	DSLSMSC-Swap	90.20±0.00	53.51±0.00	85.65±0.00	91.79±0.00	62.84±0.00	80.28±0.00
	DSLSMSC	**97.72±0.00**	**83.42±0.00**	**96.54±0.00**	**98.42±0.00**	**90.25±0.00**	**94.73±0.00**
Caltech101-7	DSLSMSC-L/L	66.24±0.47	75.47±0.14	68.37±0.36	96.44±0.01	56.64±0.41	52.66±0.42
	DSLSMSC-S/S	65.20±0.00	75.27±0.00	67.57±0.00	96.40±0.00	55.73±0.00	51.72±0.00
	DSLSMSC-Swap	65.20±0.00	75.27±0.00	67.57±0.00	96.40±0.00	55.73±0.00	51.72±0.00
	DSLSMSC	**67.35±0.07**	**75.49±0.17**	**68.67±0.09**	**97.09±0.03**	**56.92±0.11**	**53.12±0.09**
CiteSeer	DSLSMSC-L/L	53.41±0.00	24.44±0.00	37.22±0.00	37.46±0.00	23.68±0.00	36.98±0.00
	DSLSMSC-S/S	51.33±0.02	25.97±0.05	36.29±0.02	36.19±0.01	22.49±0.02	36.38±0.02
	DSLSMSC-Swap	49.90±0.04	22.32±0.03	34.19±0.04	34.73±0.04	20.15±0.04	33.67±0.04
	DSLSMSC	**61.85±0.02**	**35.73±0.03**	**45.64±0.02**	**45.55±0.02**	**33.80±0.02**	**45.73±0.02**



(a) ORL (b) MSRC-V1 (c) 100leaves

(d) BBC (e) BBCSport (f) WebKB

Fig. 6. Convergence results of DSLSMSC.

Table 15
T_F values corresponding to each metric.

Metrics	T_F
ACC	14.3837
NMI	17.0377
F-score	12.6377
Precision	14.5361
AR	14.4624
Recall	9.1906

average rank across all six evaluation metrics. Overall, these results further confirm the statistical reliability of the performance advantage of DSLSMSC.

5.3.1. Convergence analysis

This section investigates the convergence properties of the proposed DSLSMSC model by evaluating its iteration error curves across six benchmark datasets: ORL, MSRC-V1, 100leaves, BBC, BBCSport, and WebKB. The convergence curves depicted in Fig. 6 are derived from Eq. (30), where r_1, r_2, r_3 and r_4 denote reconstruction errors, while m_1, m_2, m_3 and m_4 represent matching errors. As illustrated in Fig. 6, DSLSMSC exhibits a highly significant convergence trend as iterations progress, typically stabilizing within 40 to 50 iterations. This behavior indicates that the optimization reaches a stable solution under the reported experimental settings.

6. Conclusion

This paper presents DSLSMSC, which learns latent self representations from a joint view space and multiple original view spaces. Low-rank regularization is applied to the joint view self representation to capture shared global structure, whereas sparse regularization is applied to the original view self representations to retain local relationships within individual views. TNN regularization integrates these representations and captures their high-order correlations. An ADMM based algorithm is developed to optimize the resulting objective function Experimental results on seven benchmark datasets show that DSLSMSC achieves favorable clustering results. Nevertheless, the proposed method still has some limitations. The dual-space optimization procedure together with tensor regularization introduces relatively high computational cost, which may limit scalability in large-scale scenarios. In addition, clustering performance is influenced by the latent dimension k, indicating that the choice of latent dimension remains important in practical applications. Future work will therefore focus on improving computational efficiency and exploring more adaptive parameter selection strategies to reduce reliance on manual tuning.

CRediT authorship contribution statement

Lihao Yang: Writing – original draft, Conceptualization; **Yong Wang:** Writing – review & editing, Supervision; **Yourui Huang:** Writing – review & editing, Validation, Conceptualization; **Guifu Lu:** Writing – review & editing, Conceptualization; **Geng Huang:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

Brbić, M., & Kopriva, I. (2018). Multi-view low-rank sparse subspace clustering. *Pattern Recognition*, *73*, 247–258.

Cai, B., Lu, G.-F., Ji, G., & Du, Y. (2025). Tensor multi-subspace learning for robust tensor-based multi-view clustering. *Knowledge-Based Systems*, *318*, 113476.

Cai, H., Wang, Y., Qi, F., Wang, Z., & Cheung, Y.-m. (2024). Multiview tensor spectral clustering via co-regularization. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *46*(10), 6795–6808.

Chao, G., Sun, S., & Bi, J. (2021). A survey on multiview clustering. *IEEE Transactions on Artificial Intelligence*, *2*(2), 146–168.

Chen, J., Yang, S., Mao, H., & Fahy, C. (2021). Multiview subspace clustering using low-rank representation. *IEEE Transactions on Cybernetics*, *52*(11), 12364–12378.

Chen, M.-S., Huang, L., Wang, C.-D., & Huang, D. (2020). Multi-view clustering in latent embedding space. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 3513–3520). (*vol. 34*).

Cui, J., Li, Y., Huang, H., & Wen, J. (2024). Dual contrast-driven deep multi-view clustering. *IEEE Transactions on Image Processing*, *33*, 4753–4764.

Fu, Z., Zhao, Y., Chang, D., Zhang, X., & Wang, Y. (2022). Auto-weighted low-rank representation for clustering. *Knowledge-Based Systems*, *251*, 109063.

Gao, J., Liu, M., Li, P., Zhang, J., & Chen, Z. (2023). Deep multiview adaptive clustering with semantic invariance. *IEEE Transactions on Neural Networks and Learning Systems*, *35*(9), 12965–12978.

He, H., Cai, B., & Wang, X. (2025). Multi-view subspace clustering with a consensus tensorized scaled simplex representation. *Information Sciences*, *695*, 121730.

Huang, S., Liu, Y., Tsang, I. W., Xu, Z., & Lv, J. (2022). Multi-view subspace clustering by joint measuring of consistency and diversity. *IEEE Transactions on Knowledge and Data Engineering*, *35*(8), 8270–8281.

Ji, J., & Feng, S. (2025). Partition-level fusion induced multi-view subspace clustering with tensorial geman rank. *Neural Networks*, *182*, 106849.

Kilmer, M. E., & Martin, C. D. (2011). Factorization strategies for third-order tensors. *Linear Algebra and its Applications*, *435*(3), 641–658.

Lan, S., Zheng, Q., & Yu, Y. (2024). Double-level view-correlation multi-view subspace clustering. *Knowledge-Based Systems*, *284*, 111271.

Li, J., Gao, Q., Wang, Q., Yang, M., & Xia, W. (2023). Orthogonal non-negative tensor factorization based multi-view clustering. *Advances in neural information processing systems*, *vol. 36*, 18186–18202.

Li, S., Yang, C., & Guo, H. (2025). Auto-adjustable dual-information graph regularized NMF for multiview data clustering. *Pattern Recognition*, *166*, 111679.

Lin, Z., Liu, R., & Su, Z. (2011). Linearized alternating direction method with adaptive penalty for low-rank representation. *Advances in neural information processing systems*, *vol. 24*.

Liu, G., Lin, Z., Yan, S., Sun, J., Yu, Y., & Ma, Y. (2012). Robust recovery of subspace structures by low-rank representation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *35*(1), 171–184.

Liu, J., Wang, C., Gao, J., & Han, J. (2013). Multi-view clustering via joint nonnegative matrix factorization. In *Proceedings of the 2013 SIAM international conference on data mining* (pp. 252–260). SIAM.

Luo, Y., Ma, W., Liu, S., Tong, X., Du, X., Liu, Z., Li, M., & Jiang, H. (2025). Multi-view subspace clustering via slack consistency and double-side orthogonal diversity. *Expert Systems with Applications*, *277*, 127118.

MacQueen, J. (1967). Some methods for classification and analysis of multivariate observations. In *Proceedings of the fifth berkeley symposium on mathematical statistics and probability, volume 1: Statistics* (pp. 281–298). University of California press (*vol. 5*).

Nie, F., Tian, L., & Li, X. (2018). Multiview clustering via adaptively weighted procrustes. In *Proceedings of the 24th ACM SIGKDD international conference on knowledge discovery & data mining* (pp. 2022–2030).

Sun, W., Li, C., Li, Q., Fang, X., He, J., & Liu, L. (2025). Joint intra-view and inter-view enhanced tensor low-rank induced affinity graph learning. *Pattern Recognition*, *159*, 111140.

Tao, Q., Tonin, F., Patrinos, P., & Suykens, J. A. K. (2024). Tensor-based multi-view spectral clustering via shared latent space. *Information Fusion*, *108*, 102405.

Wang, S., Chen, Y., Jin, Y., Cen, Y., Li, Y., & Zhang, L. (2021). Error-robust low-rank tensor approximation for multi-view clustering. *Knowledge-Based Systems*, *215*, 106745.

Wu, T., & Lu, G.-F. (2024). Coupled double self-expressive subspace clustering with low-rank tensor learning. *Expert Systems with Applications*, *255*, 124619.

Wu, T., & Lu, G.-F. (2025). Tensorized diversity and consistency with laplacian manifold for multi-view clustering. *Information Sciences*, *690*, 121575.

Xia, W., Zhang, X., Gao, Q., Shu, X., Han, J., & Gao, X. (2021). Multiview subspace clustering by an enhanced tensor nuclear norm. *IEEE Transactions on cybernetics*, *52*(9), 8962–8975.

Xie, D., Gao, Q., & Yang, M. (2023). Enhanced tensor low-rank representation learning for multi-view clustering. *Neural Networks*, *161*, 93–104.

Xie, D., Gao, Q., Zhao, Y., Yang, F., & Song, W. (2024). Consistent graph learning for multi-view spectral clustering. *Pattern Recognition*, *154*, 110598.

Xie, Y., Tao, D., Zhang, W., Liu, Y., Zhang, L., & Qu, Y. (2018). On unifying multi-view self-representations for clustering by tensor multi-rank minimization. *International Journal of Computer Vision*, *126*, 1157–1179.

Yu, X., Liu, H., Zhang, Y., Gao, Y., & Zhang, C. (2025). Robust multi-view clustering with hyper-laplacian regularization. *Information Sciences*, *694*, 121718.

Zhang, C., Fu, H., Liu, S., Liu, G., & Cao, X. (2015). Low-rank tensor constrained multiview subspace clustering. In *Proceedings of the IEEE international conference on computer vision* (pp. 1582–1590).

Zhang, C., Hu, Q., Fu, H., Zhu, P., & Cao, X. (2017). Latent multi-view subspace clustering. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 4279–4287).

Zhang, G.-Y., Huang, D., & Wang, C.-D. (2023). Facilitated low-rank multi-view subspace clustering. *Knowledge-Based Systems*, *260*, 110141.
Zhao, J.-B., & Lu, G.-F. (2022). Clean and robust affinity matrix learning for multi-view clustering. *Applied Intelligence*, *52*(14), 15899–15915.
Zheng, Q., Zhu, J., Li, Z., Pang, S., Wang, J., & Li, Y. (2020a). Feature concatenation multi-view subspace clustering. *Neurocomputing*, *379*, 89–102.
Zheng, Q., Zhu, J., Li, Z., Tian, Z., & Li, C. (2023). Comprehensive multi-view representation learning. *Information Fusion*, *89*, 198–209.
Zheng, Q., Zhu, J., Tian, Z., Li, Z., Pang, S., & Jia, X. (2020b). Constrained bilinear factorization multi-view subspace clustering. *Knowledge-Based Systems*, *194*, 105514.